



Chapter 22

Network Layer: Delivery, Forwarding, and Routing

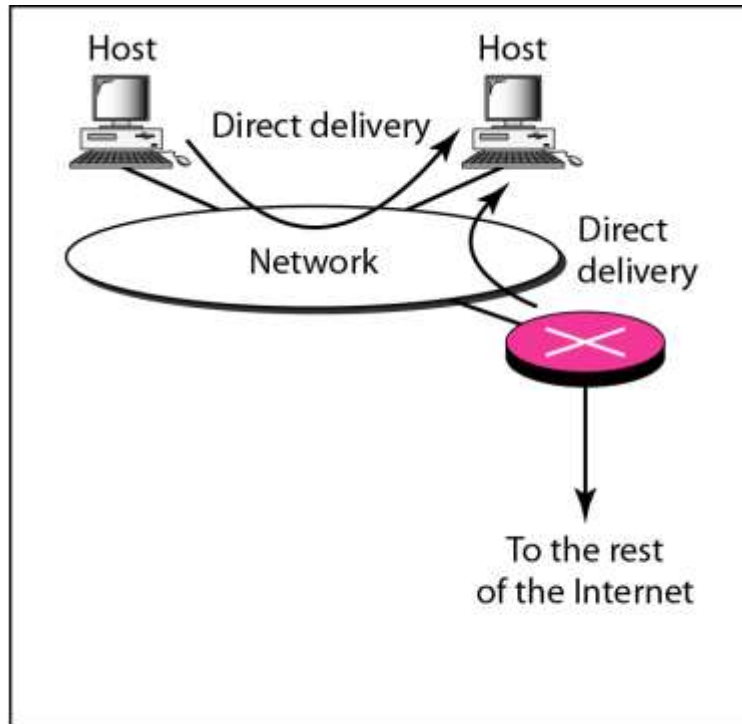
22-1 DELIVERY

The network layer supervises the handling of the packets by the underlying physical networks. We define this handling as the delivery of a packet.

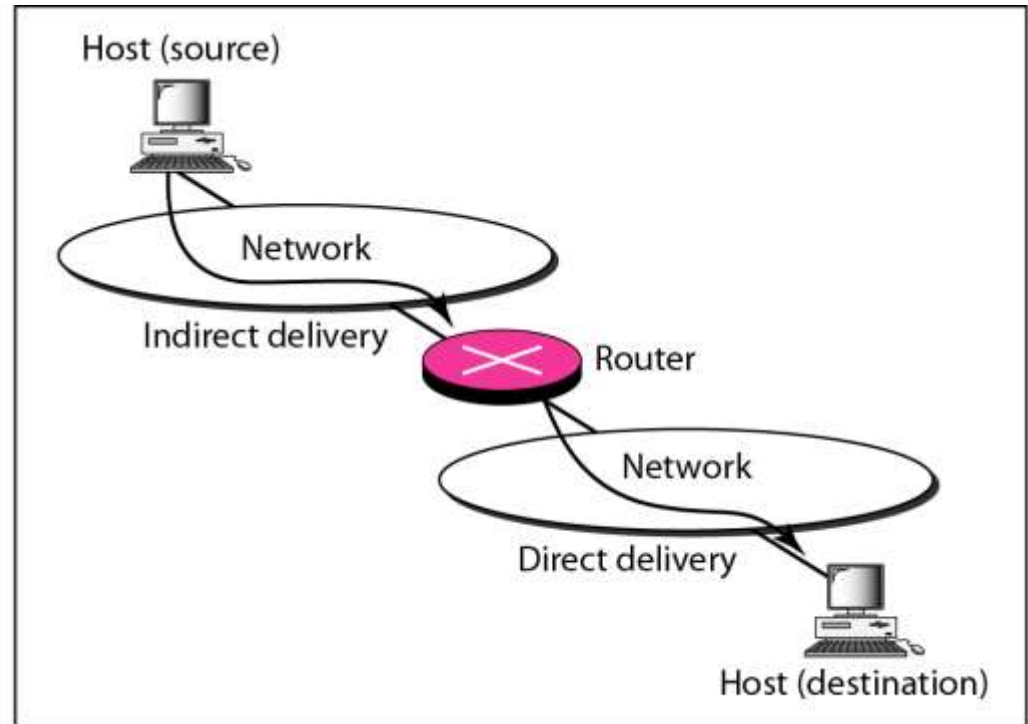
Topics discussed in this section:

Direct Versus Indirect Delivery

Figure 22.1 *Direct and indirect delivery*



a. Direct delivery



b. Indirect and direct delivery

- **Direct delivery** The router is not used to forward the packet because the destination is on the same network (subnet or network segment) as the sending host.
- **Indirect delivery** Packets are sent to an intermediate point for forwarding, usually a router.

22-2 FORWARDING

Forwarding means to place the packet in its route to its destination. Forwarding requires a host or a router to have a routing table. When a host has a packet to send or when a router has received a packet to be forwarded, it looks at this table to find the route to the final destination.

Topics discussed in this section:

Forwarding Techniques

Forwarding Process

Routing Table

Figure 22.2 *Route method versus next-hop method*

a. Routing tables based on route

Destination	Route
Host B	R1, R2, host B

Routing table
for host A

Destination	Route
Host B	R2, host B

Routing table
for R1

Destination	Route
Host B	Host B

Routing table
for R2

b. Routing tables based on next hop

Destination	Next hop
Host B	R1

Destination	Next hop
Host B	R2

Destination	Next hop
Host B	---

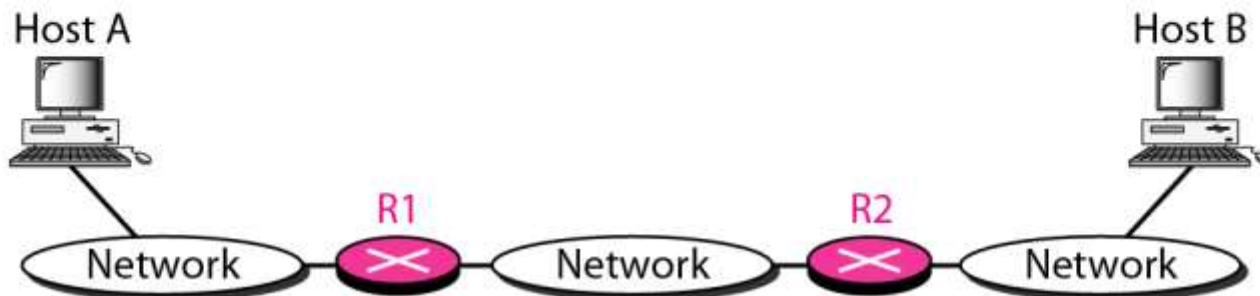


Figure 22.3 *Host-specific versus network-specific method*

Routing table for host S based on host-specific method

Destination	Next hop
A	R1
B	R1
C	R1
D	R1

Routing table for host S based on network-specific method

Destination	Next hop
N2	R1

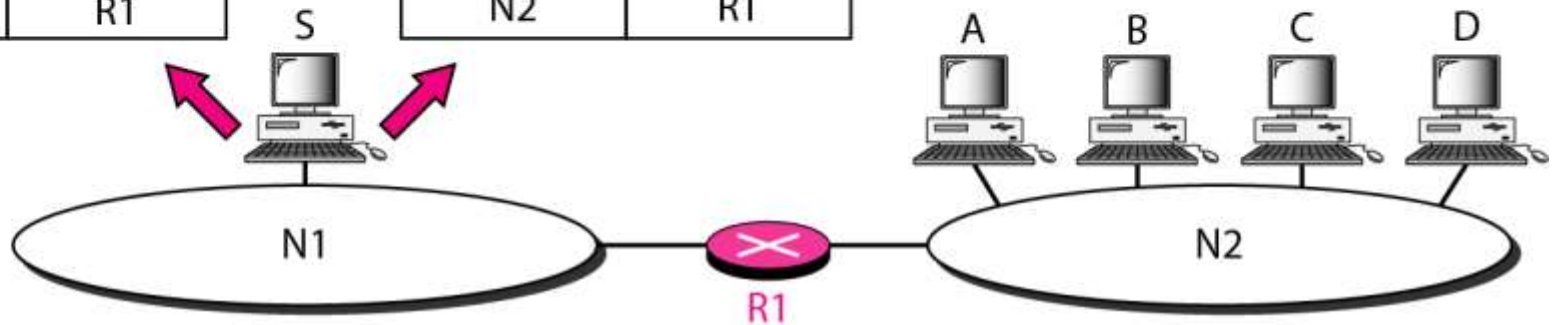
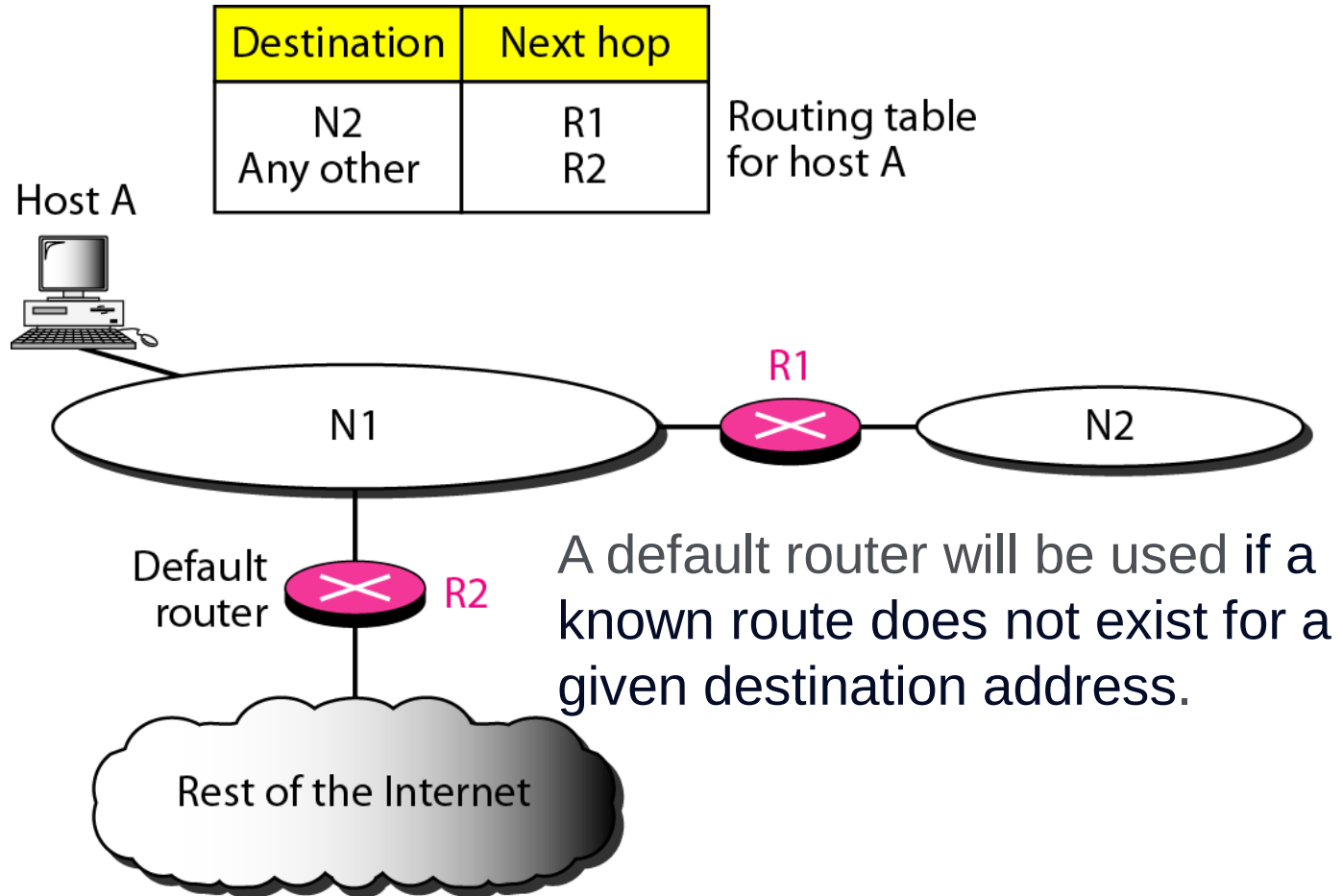


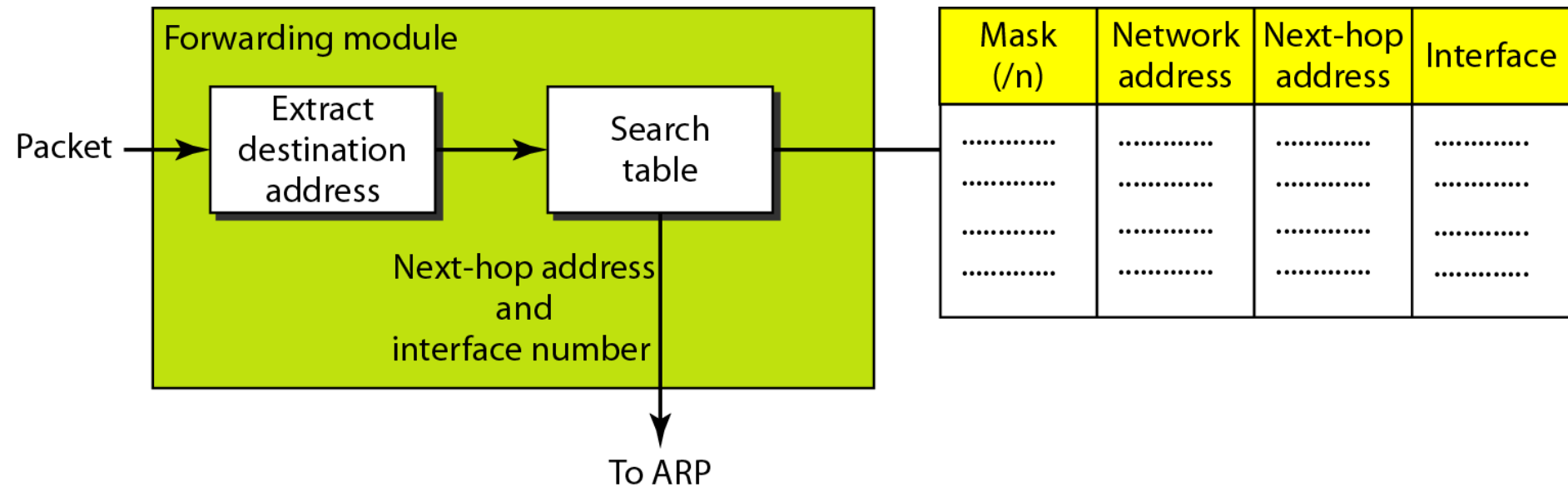
Figure 22.4 *Default method*



Subnet Mask

- A subnet mask is used to divide an IP address into two parts. One part identifies the **host** (computer), the other part identifies the **network** to which it belongs.
- A Netmask is a 32-bit "mask" used to divide an IP address into subnets and specify the network's available hosts. In a netmask, two bits are always automatically assigned. For example, in 255.255. 225.0, "0" is the assigned network address.

Figure 22.5 *Simplified forwarding module in classless address*



Note

In classless addressing, we need at least four columns in a routing table.



Example 22.1

Make a routing table for router R1, using the configuration in Figure 22.6.

Solution

Table 22.1 shows the corresponding table.

Figure 22.6 *Configuration for Example 22.1*

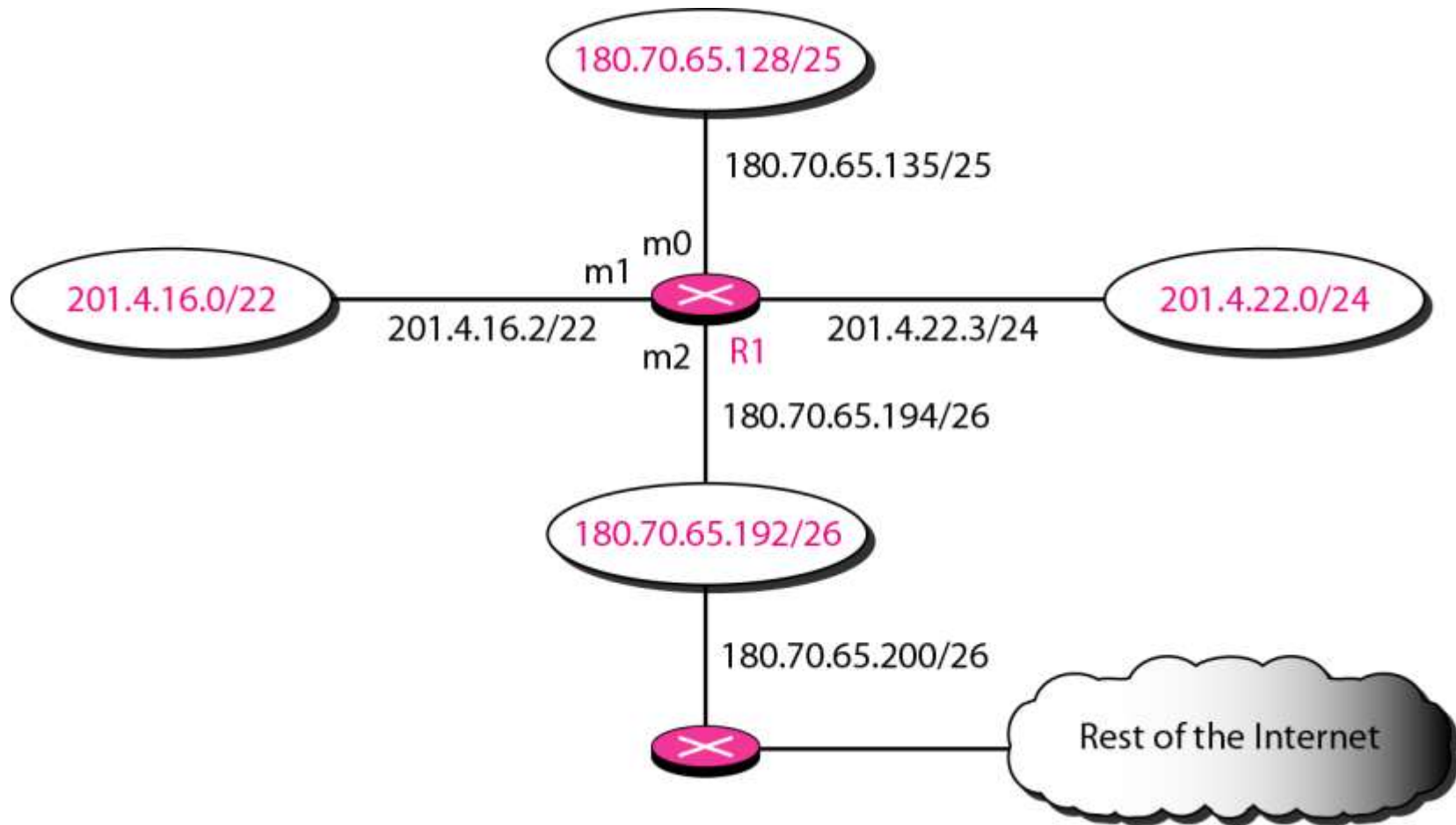


Table 22.1 *Routing table for router R1 in Figure 22.6*

<i>Mask</i>	<i>Network Address</i>	<i>Next Hop</i>	<i>Interface</i>
/26	180.70.65.192	—	m2
/25	180.70.65.128	—	m0
/24	201.4.22.0	—	m3
/22	201.4.16.0		m1
Any	Any	180.70.65.200	m2

22-3 UNICAST ROUTING PROTOCOLS

A routing table can be either static or dynamic. A static table is one with manual entries. A dynamic table is one that is updated automatically when there is a change somewhere in the Internet. A routing protocol is a combination of rules and procedures that lets routers in the Internet inform each other of changes.

Topics discussed in this section:

Optimization

Intra- and Interdomain Routing

Distance Vector Routing and RIP

Link State Routing and OSPF

Path Vector Routing and BGP

Figure 22.13 *Popular routing protocols*

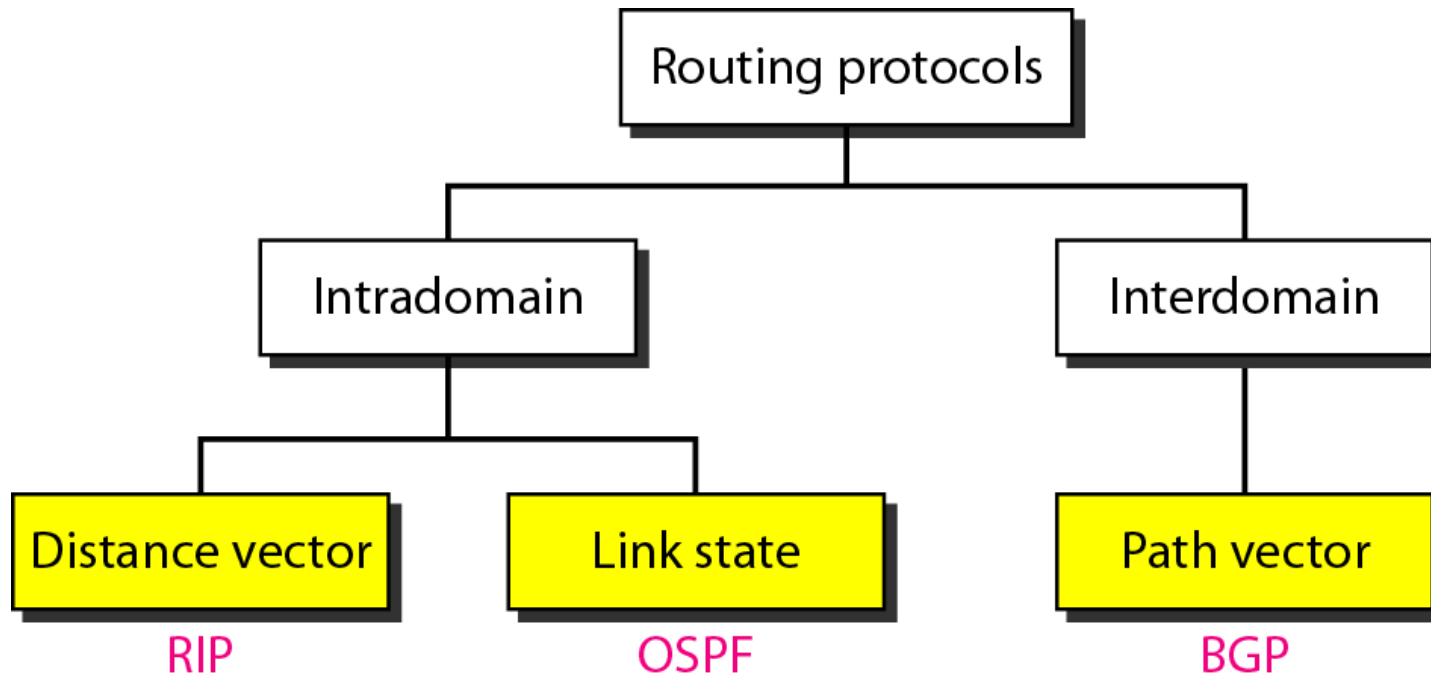


Figure 22.14 *Distance vector routing tables*

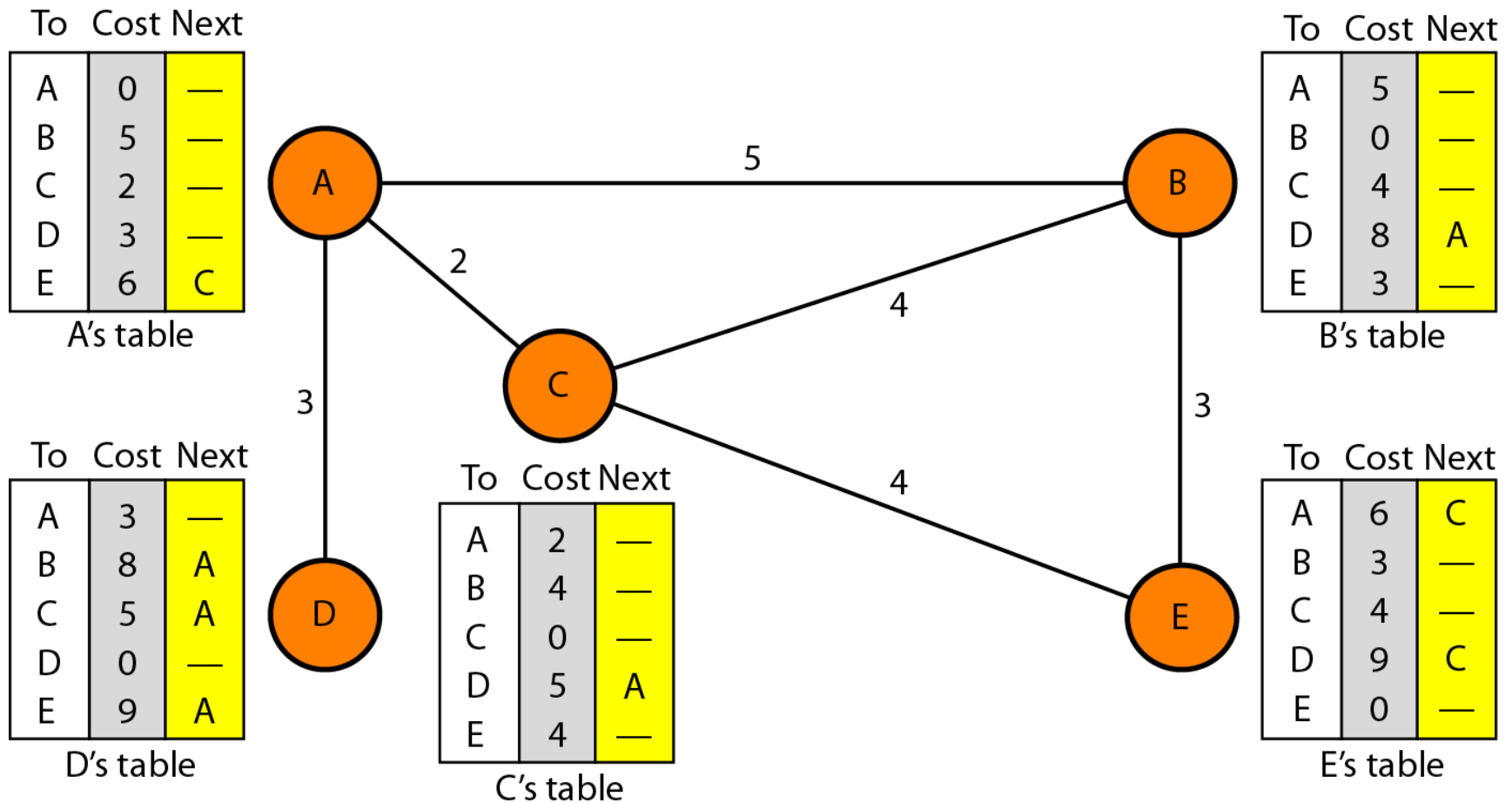
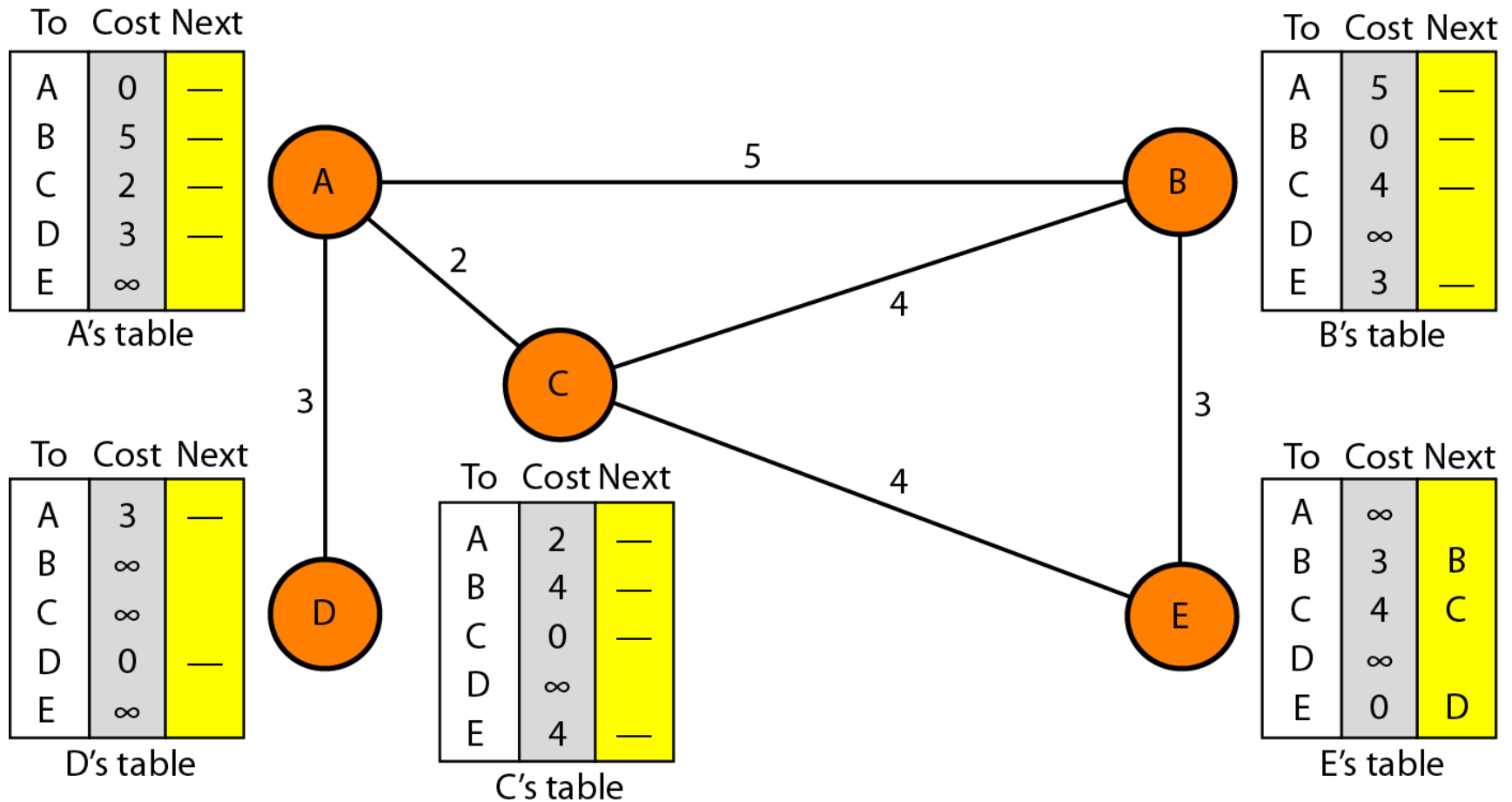


Figure 22.15 *Initialization of tables in distance vector routing*



Note

In distance vector routing, each node shares its routing table with its immediate neighbors periodically and when there is a change.

Figure 22.16 *Updating in distance vector routing*

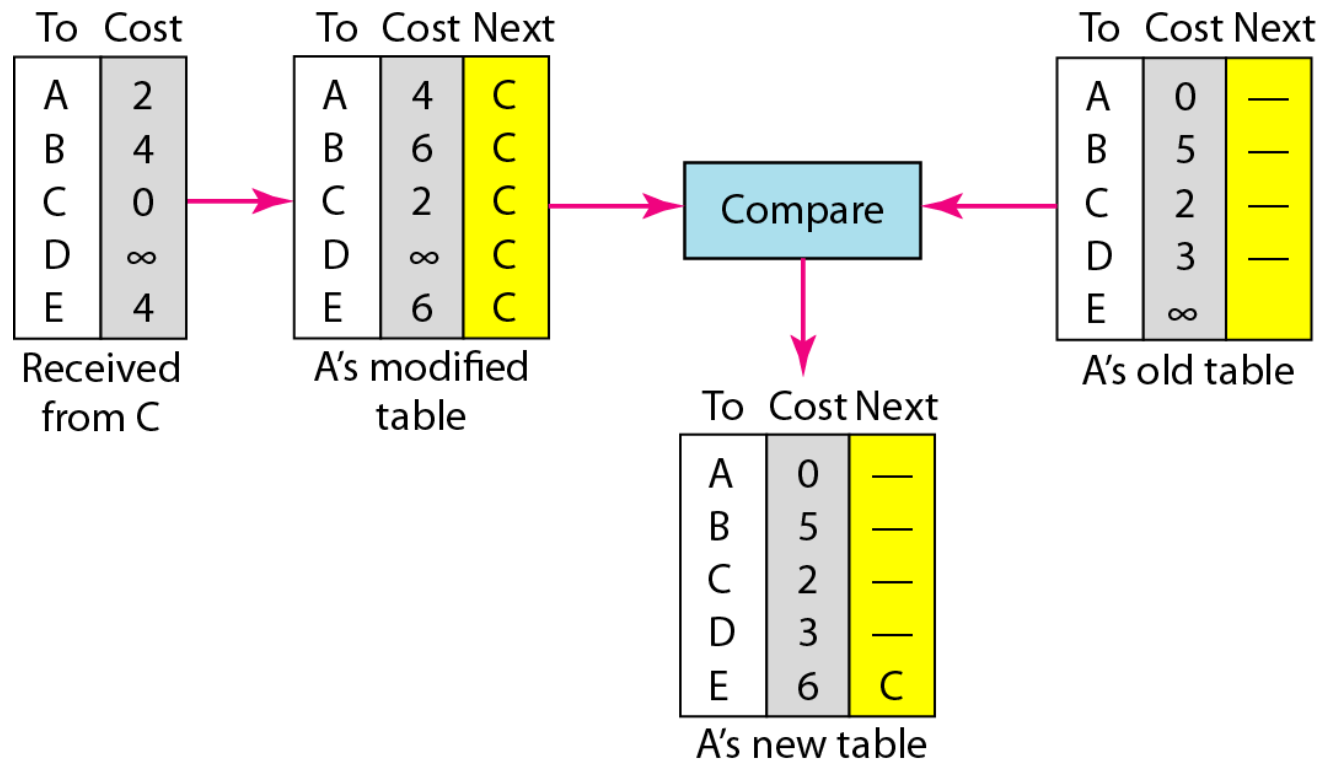


Figure 22.20 *Concept of link state routing*

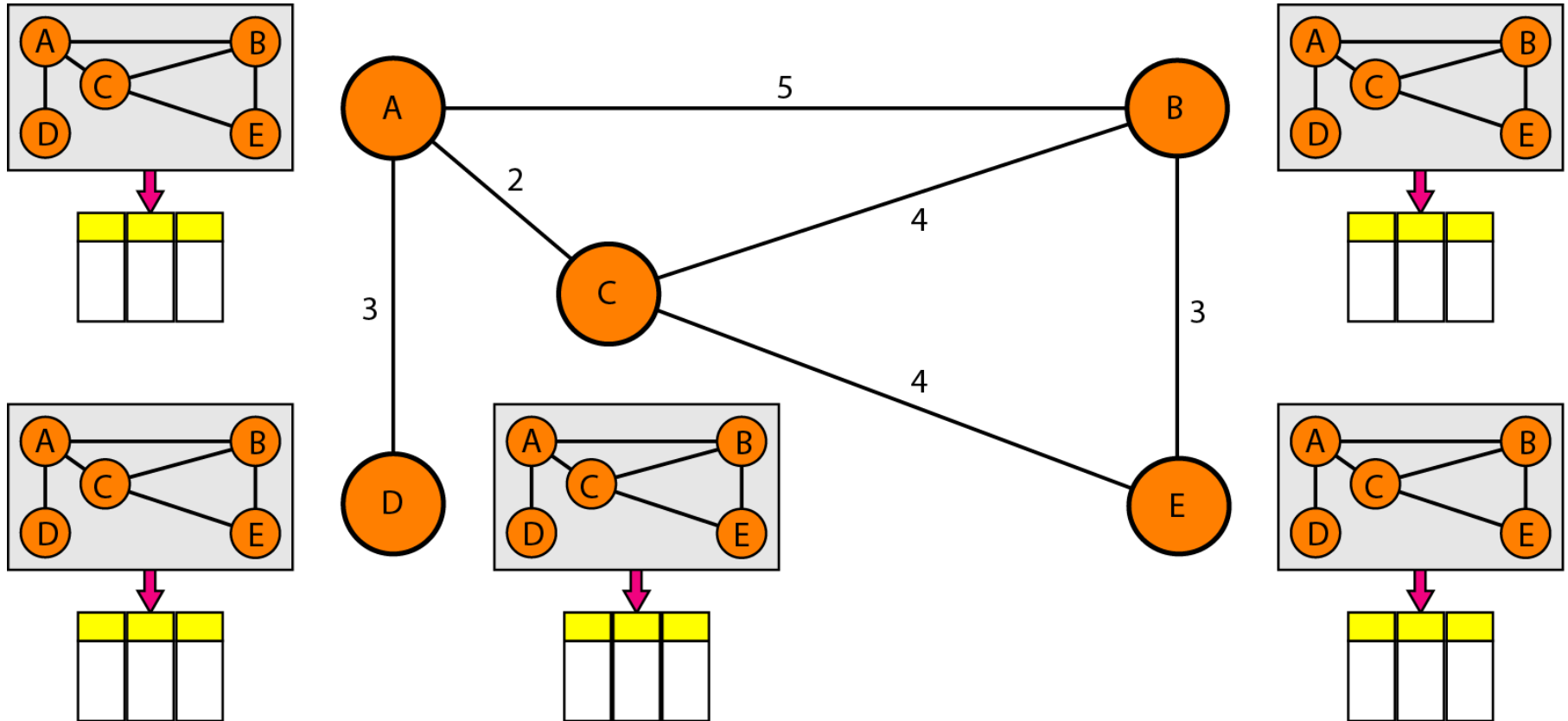


Figure 22.21 *Link state knowledge*

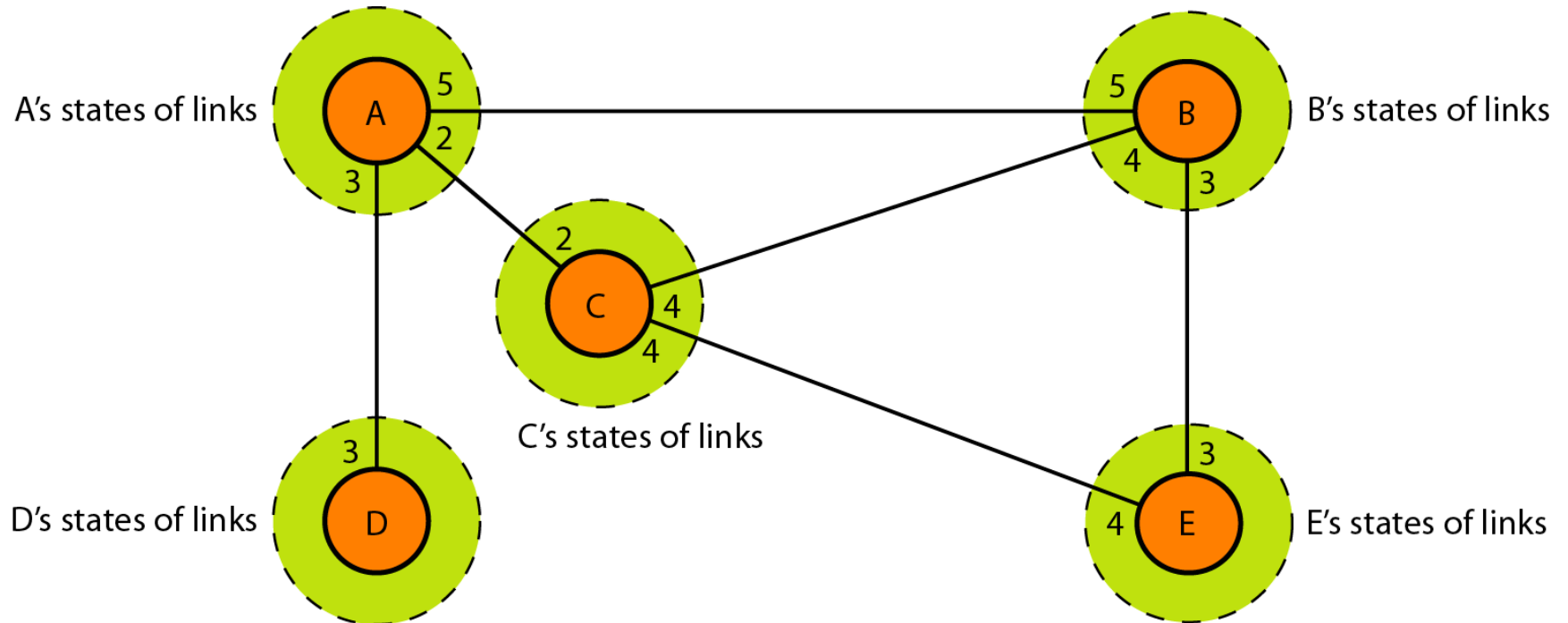


Figure 22.22 *Dijkstra algorithm*

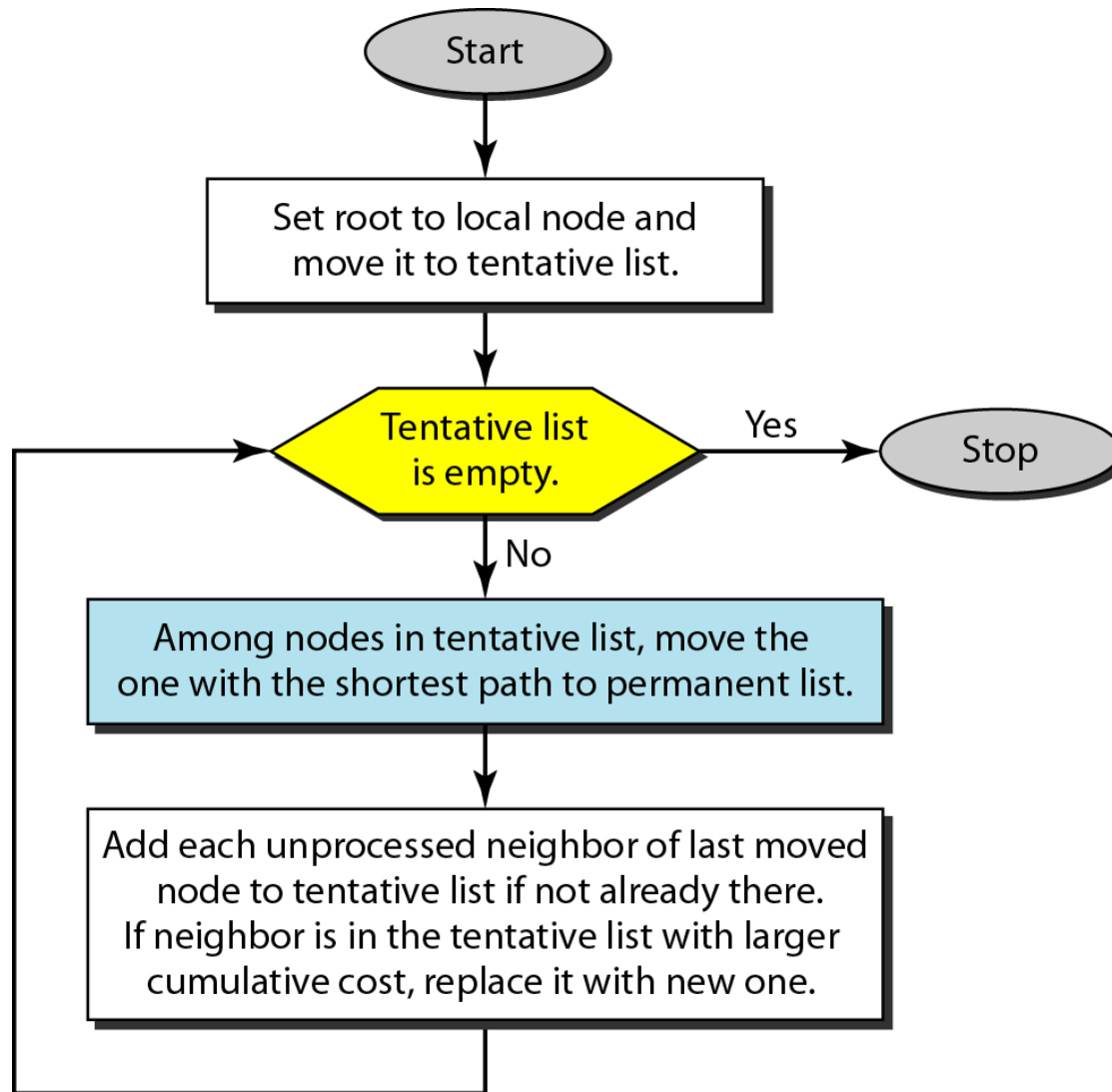


Figure 22.23 *Example of formation of shortest path tree*

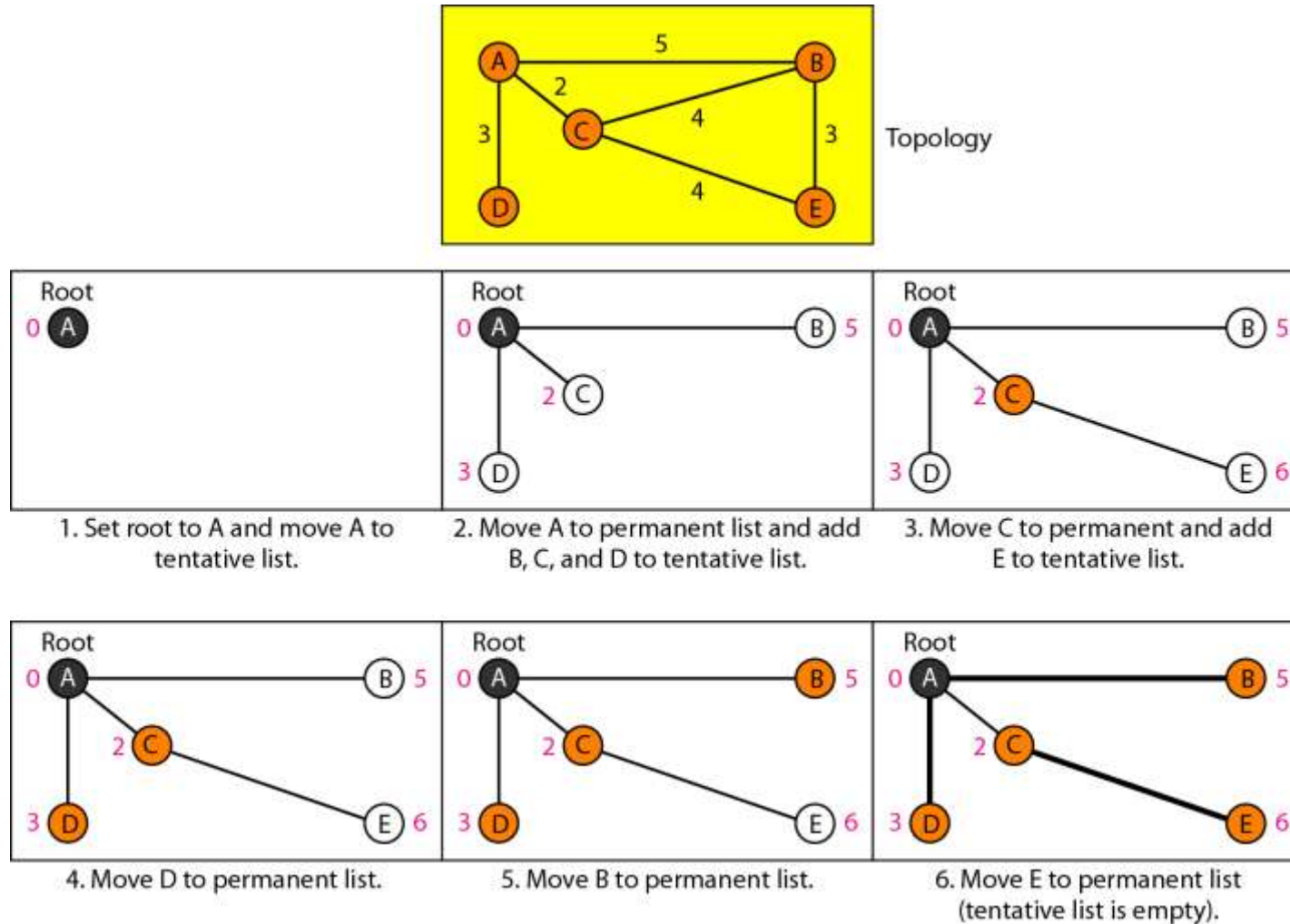


Table 22.2 *Routing table for node A*

<i>Node</i>	<i>Cost</i>	<i>Next Router</i>
A	0	—
B	5	—
C	2	—
D	3	—
E	6	C

Figure 22.24 *Areas in an autonomous system*

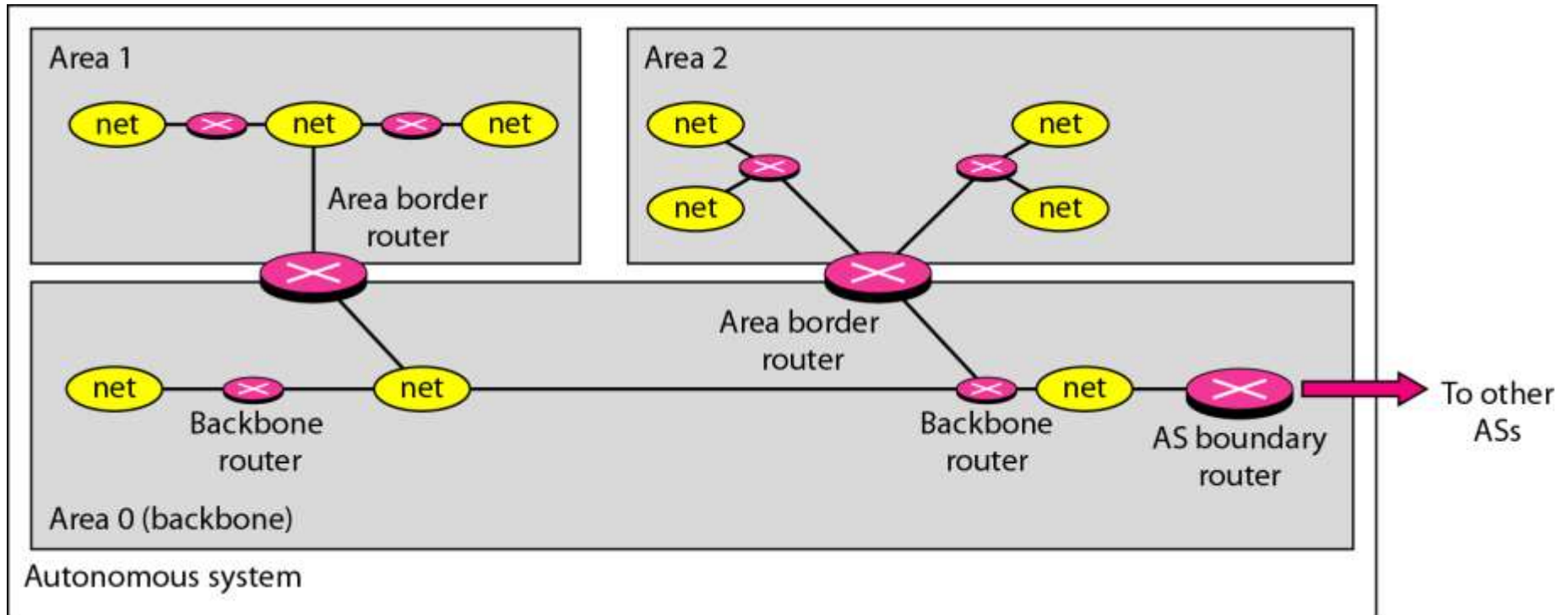


Figure 22.25 *Types of links*

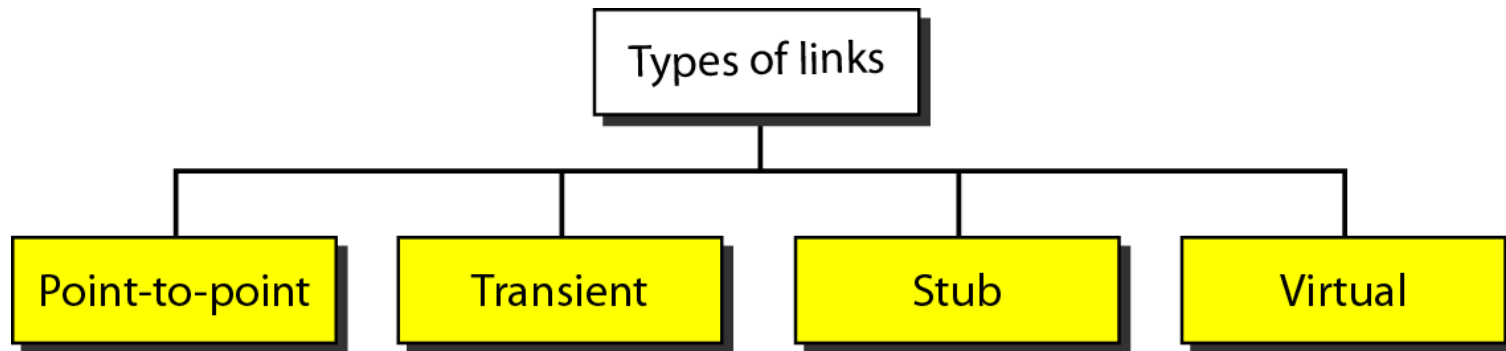


Figure 22.26 *Point-to-point link*

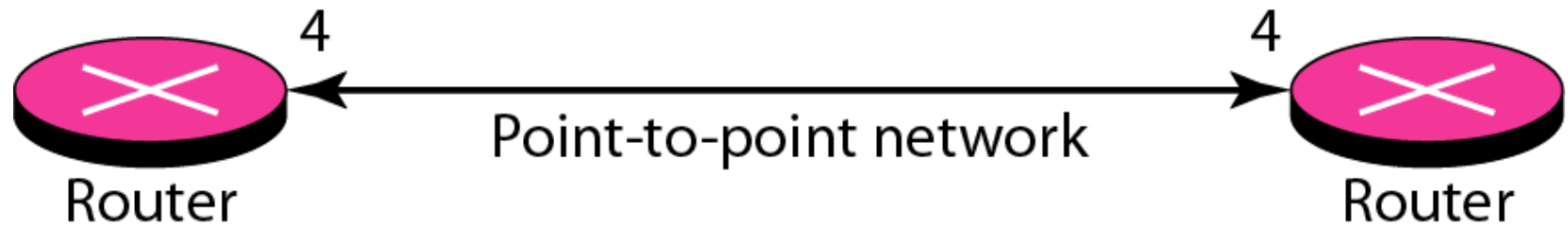
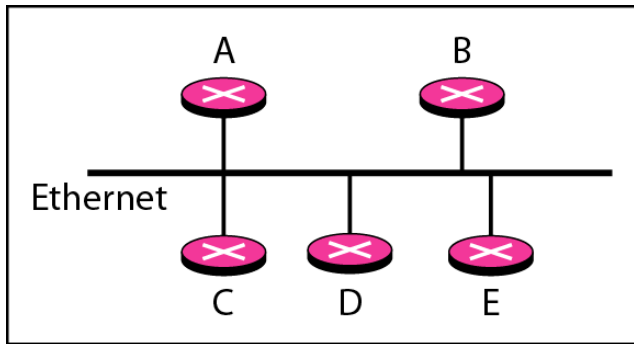
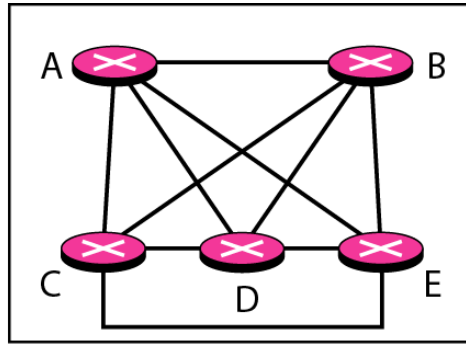


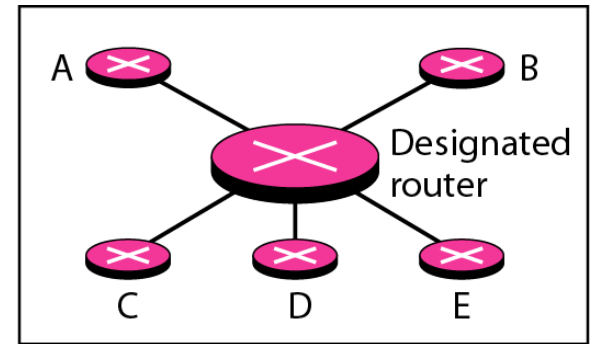
Figure 22.27 *Transient link*



a. Transient network

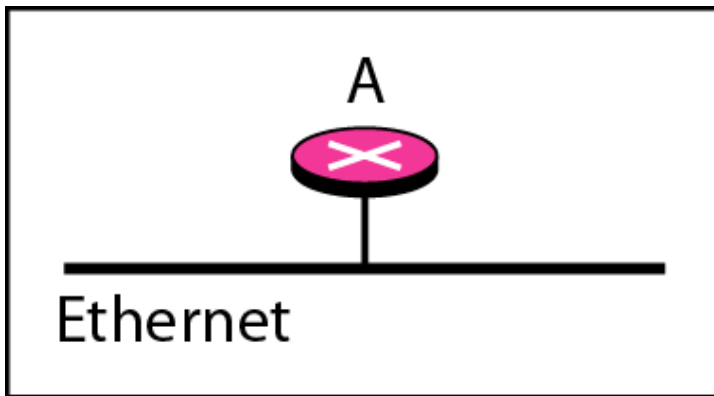


b. Unrealistic representation



c. Realistic representation

Figure 22.28 *Stub link*

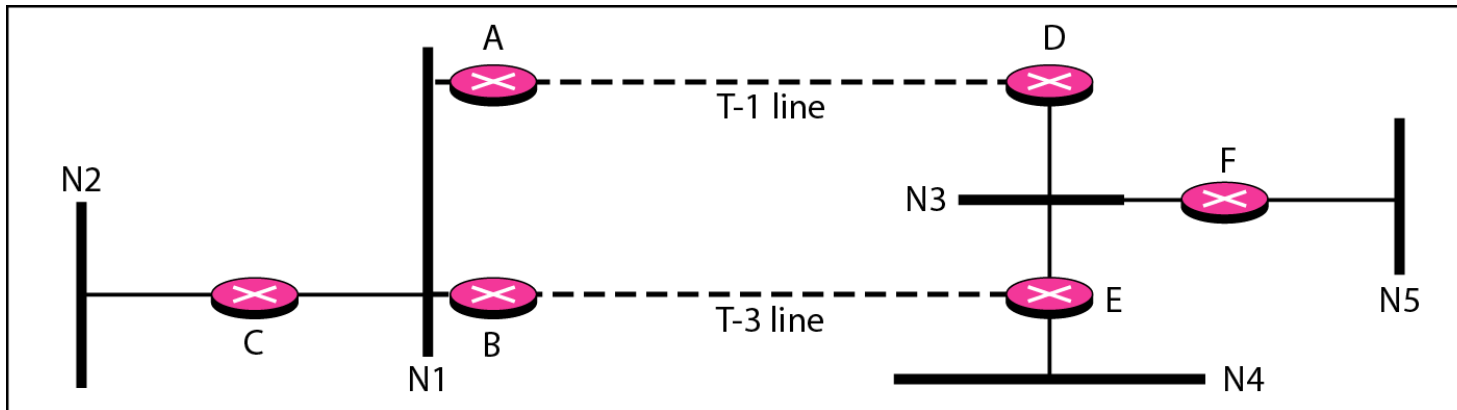


a. Stub network

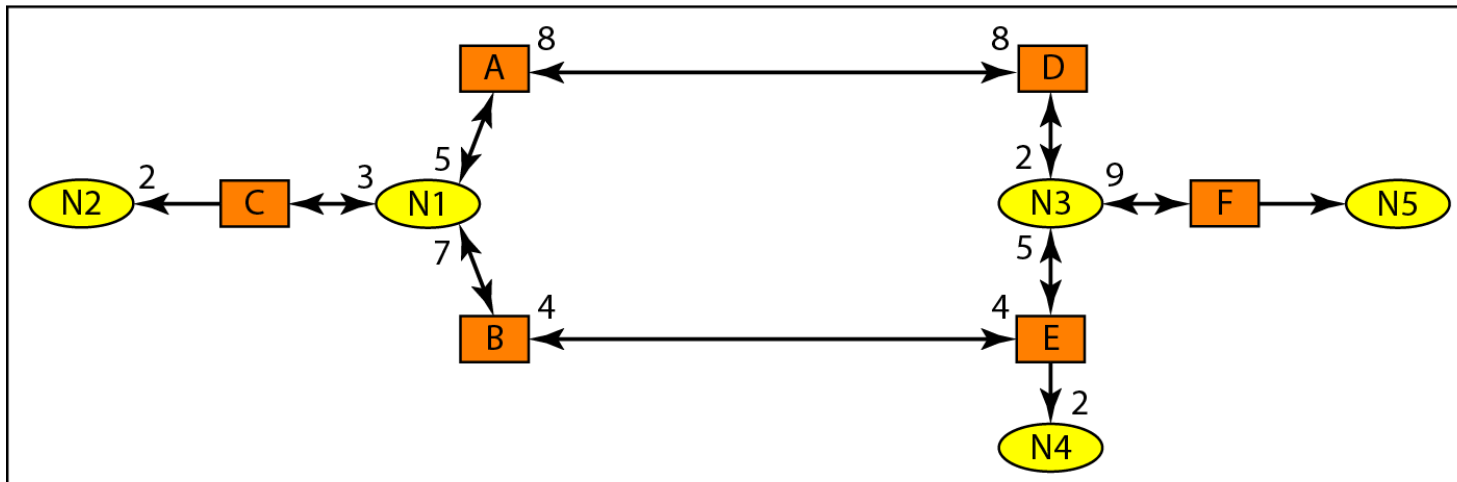


b. Representation

Figure 22.29 *Example of an AS and its graphical representation in OSPF*



a. Autonomous system



b. Graphical representation

Figure 22.30 *Initial routing tables in path vector routing*

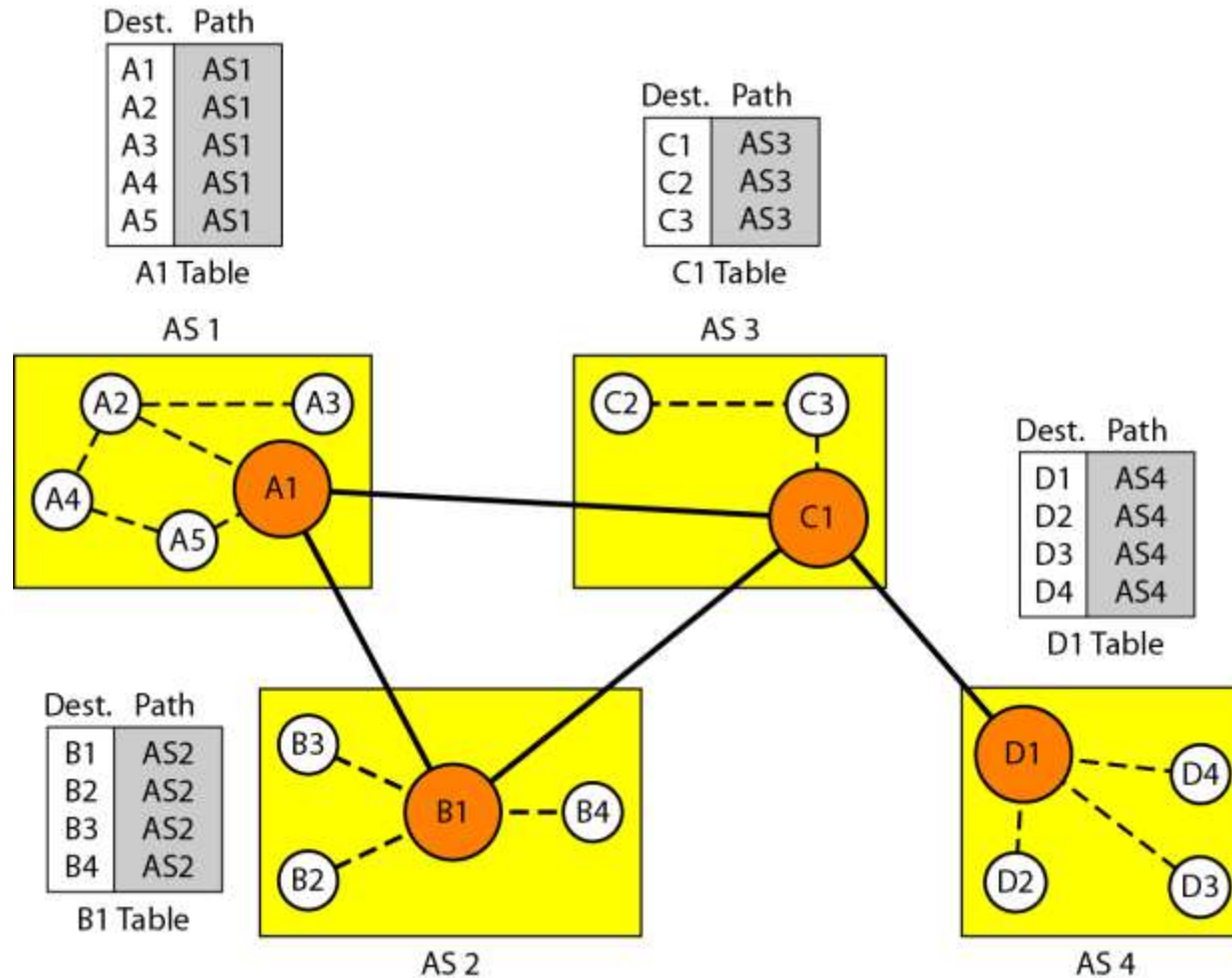


Figure 22.31 *Stabilized tables for three autonomous systems*

Dest.	Path
A1 ...	AS1
A5	AS1
B1 ...	AS1-AS2
B4	AS1-AS2
C1 ...	AS1-AS3
C3	AS1-AS3
D1 ...	AS1-AS2-AS4
D4	AS1-AS2-AS4

A1 Table

Dest.	Path
A1 ...	AS2-AS1
A5	AS2-AS1
B1 ...	AS2
B4	AS2
C1 ...	AS2-AS3
C3	AS2-AS3
D1 ...	AS2-AS3-AS4
D4	AS2-AS3-AS4

B1 Table

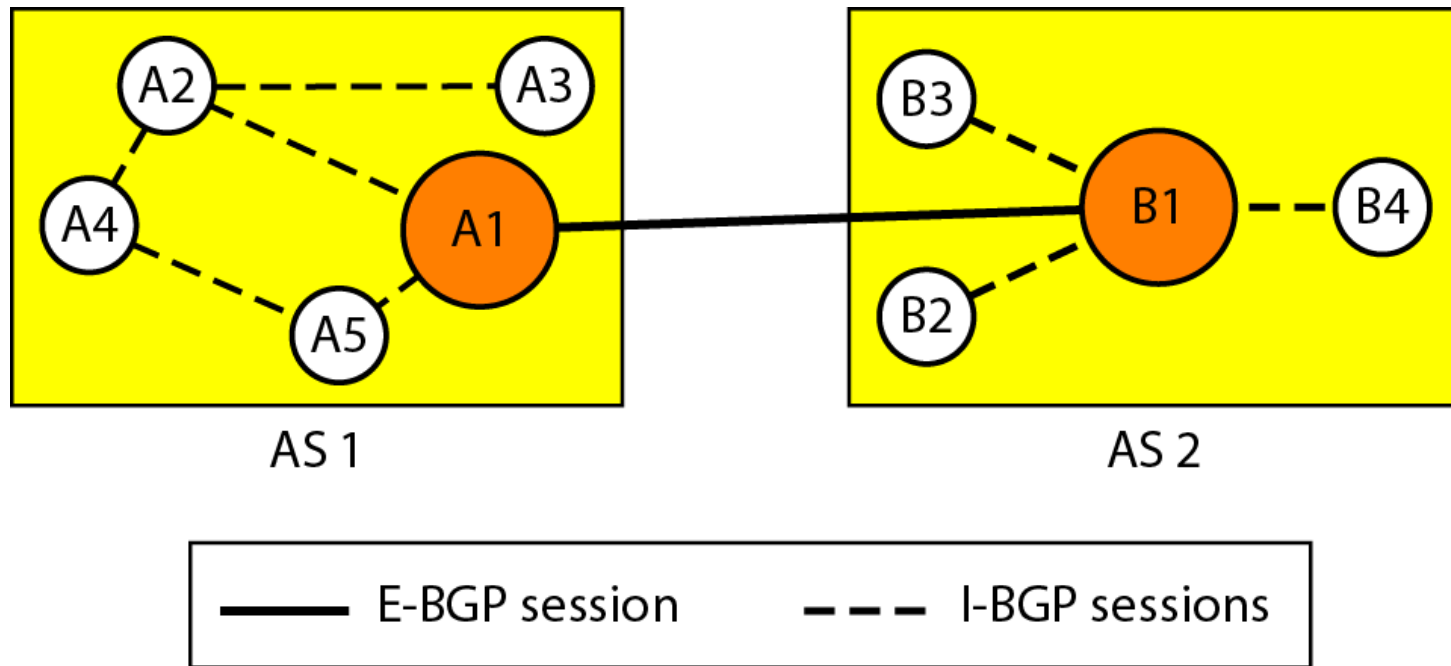
Dest.	Path
A1 ...	AS3-AS1
A5	AS3-AS1
B1 ...	AS3-AS2
B4	AS3-AS2
C1 ...	AS3
C3	AS3
D1 ...	AS3-AS4
D4	AS3-AS4

C1 Table

Dest.	Path
A1 ...	AS4-AS3-AS1
A5	AS4-AS3-AS1
B1 ...	AS4-AS3-AS2
B4	AS4-AS3-AS2
C1 ...	AS4-AS3
C3	AS4-AS3
D1 ...	AS4
D4	AS4

D1 Table

Figure 22.32 *Internal and external BGP sessions*



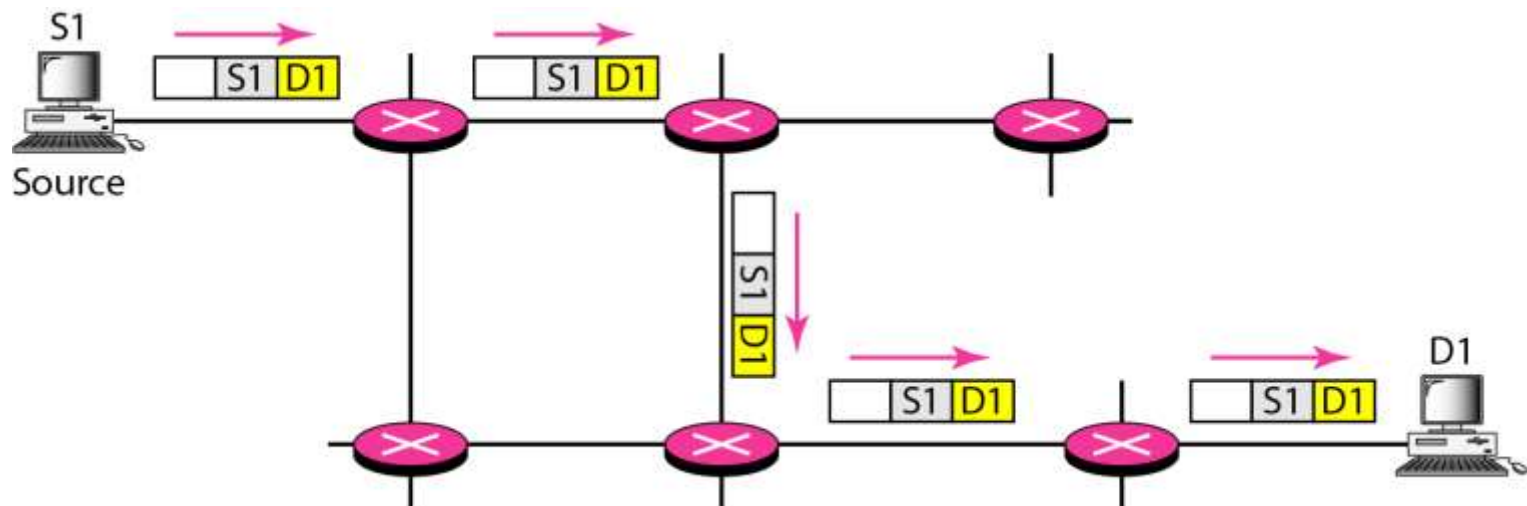
22-4 MULTICAST ROUTING PROTOCOLS

In this section, we discuss multicasting and multicast routing protocols.

Topics discussed in this section:

Unicast, Multicast, and Broadcast
Applications
Multicast Routing
Routing Protocols

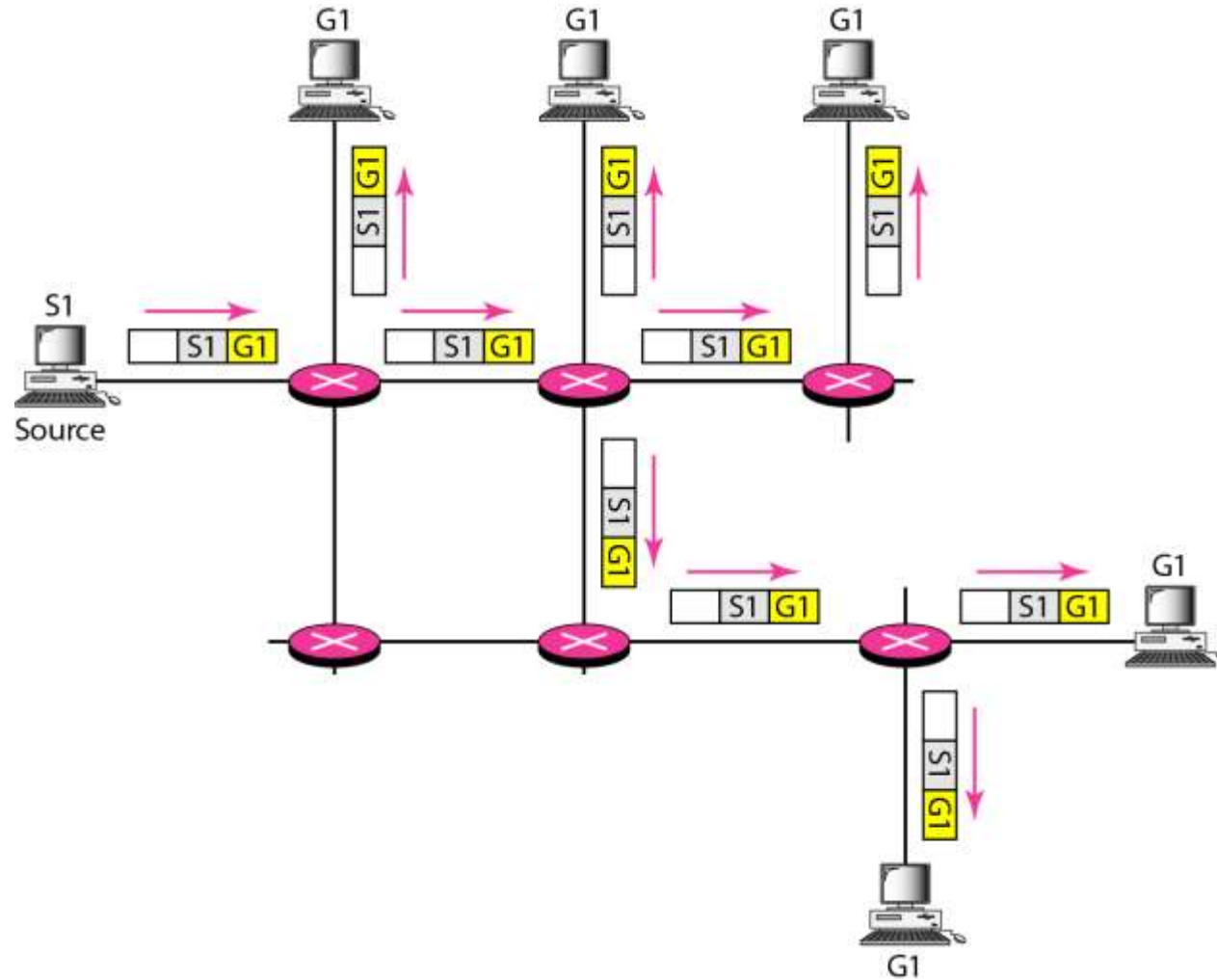
Figure 22.33 *Unicasting*



Note

In unicasting, the router forwards the received packet through only one of its interfaces.

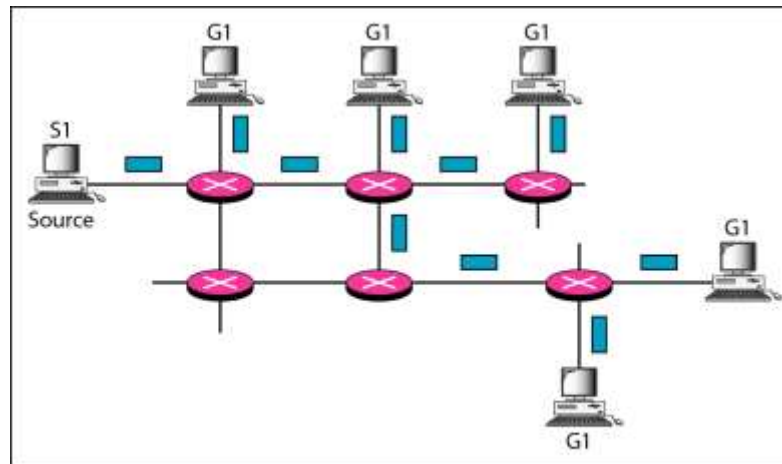
Figure 22.34 *Multicasting*



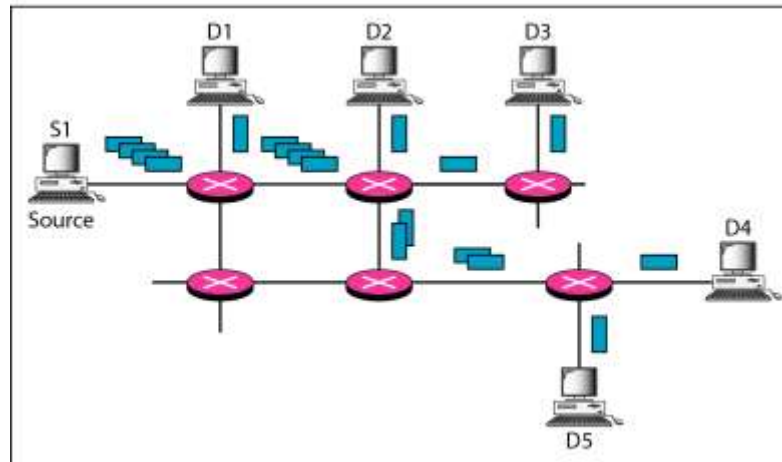
Note

In multicasting, the router may forward the received packet through several of its interfaces.

Figure 22.35 *Multicasting versus multiple unicasting*



a. Multicasting



b. Multiple unicasting

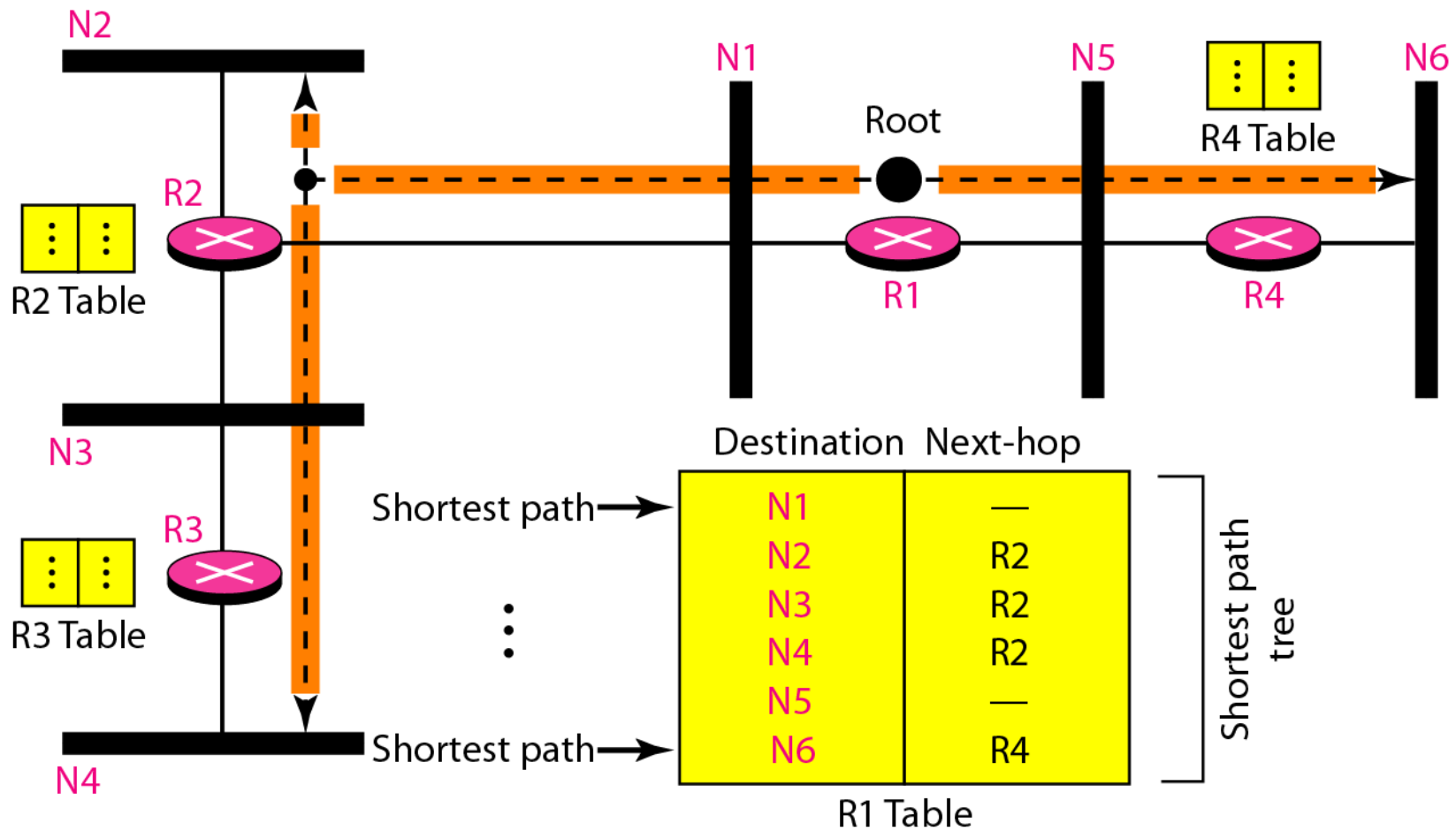
Note

Emulation of multicasting through multiple unicasting is not efficient and may create long delays, particularly with a large group.

Note

In unicast routing, each router in the domain has a table that defines a shortest path tree to possible destinations.

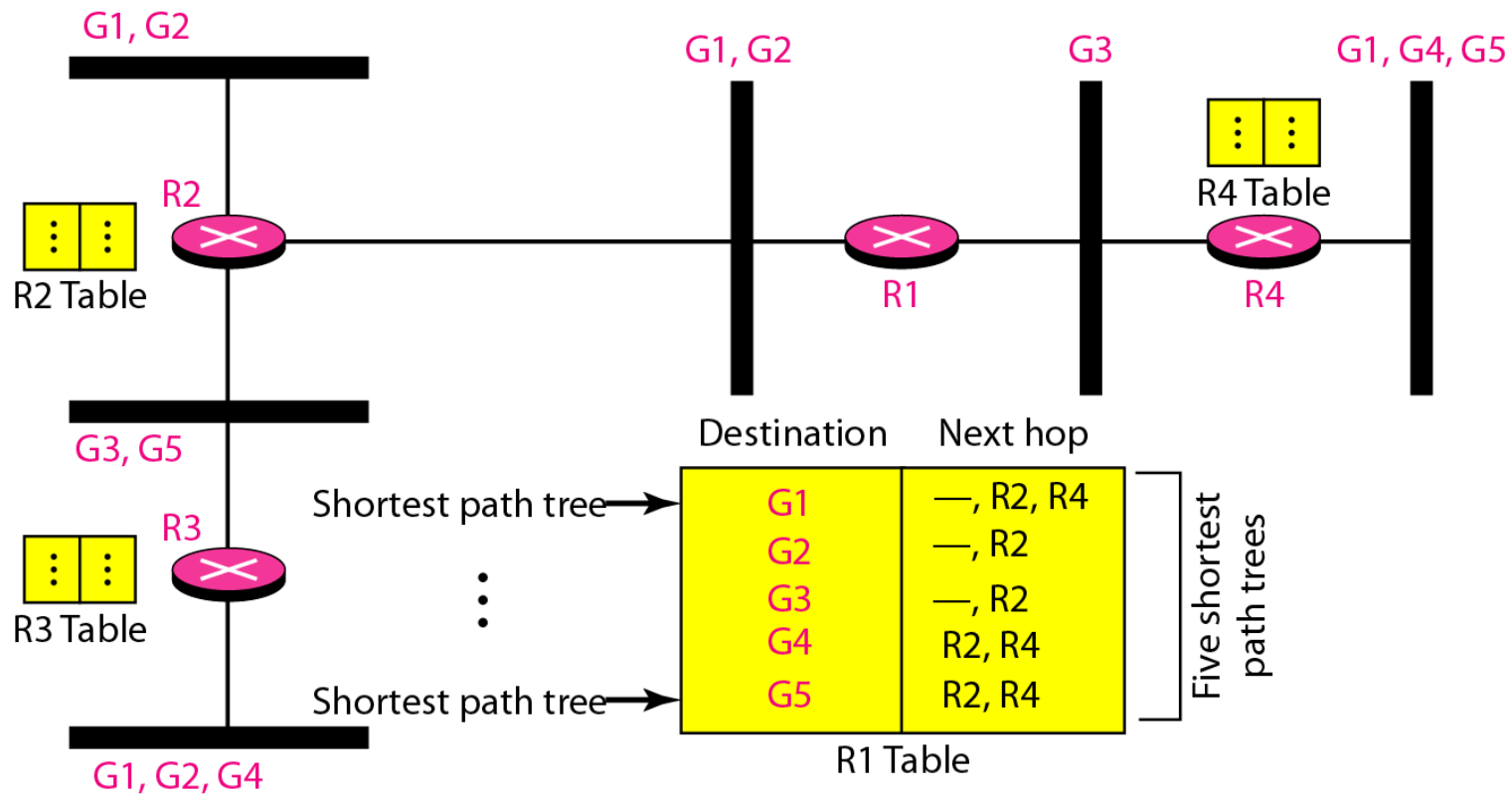
Figure 22.36 *Shortest path tree in unicast routing*



Note

In multicast routing, each involved router needs to construct a shortest path tree for each group.

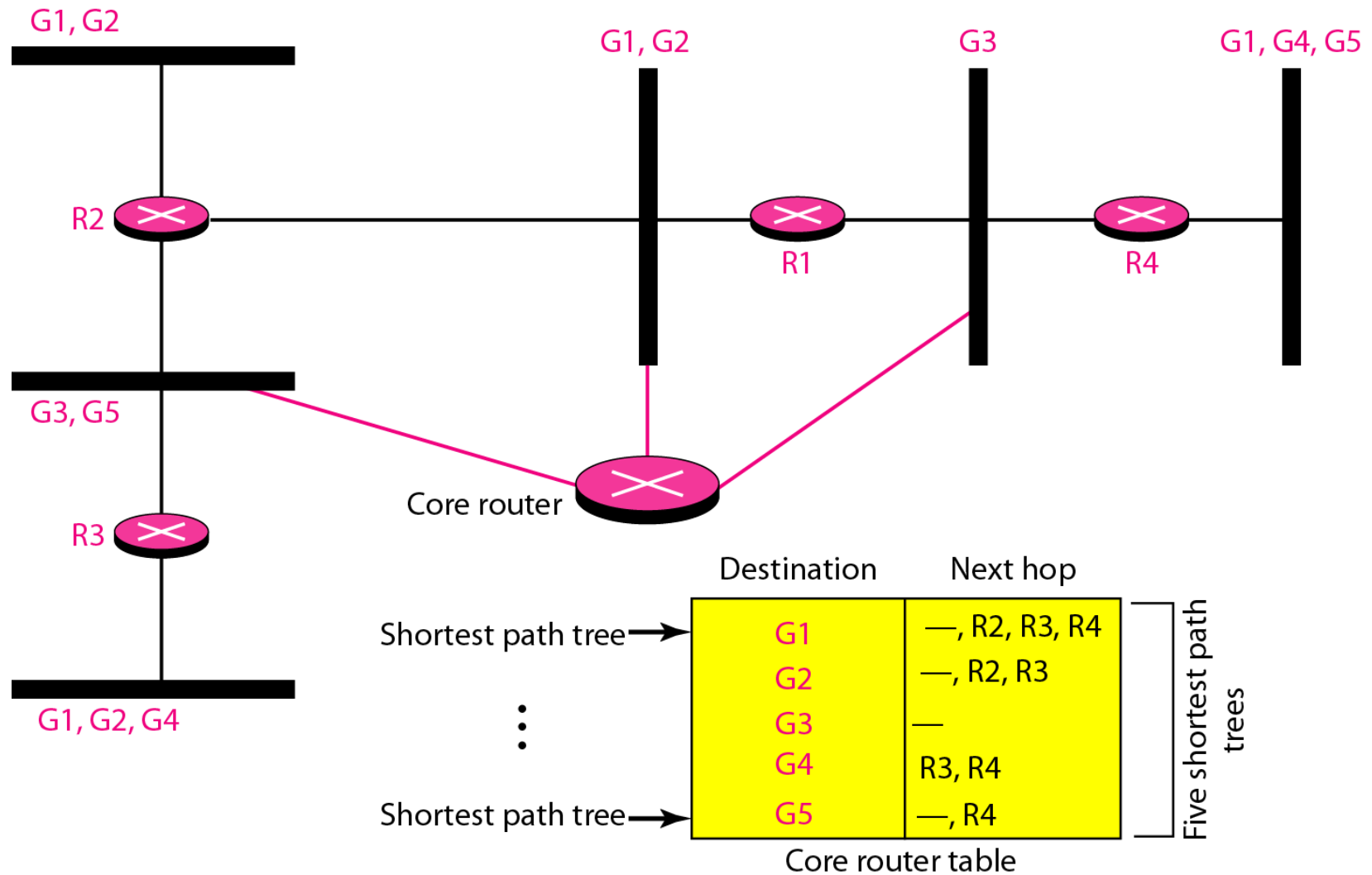
Figure 22.37 *Source-based tree approach*



Note

In the source-based tree approach, each router needs to have one shortest path tree for each group.

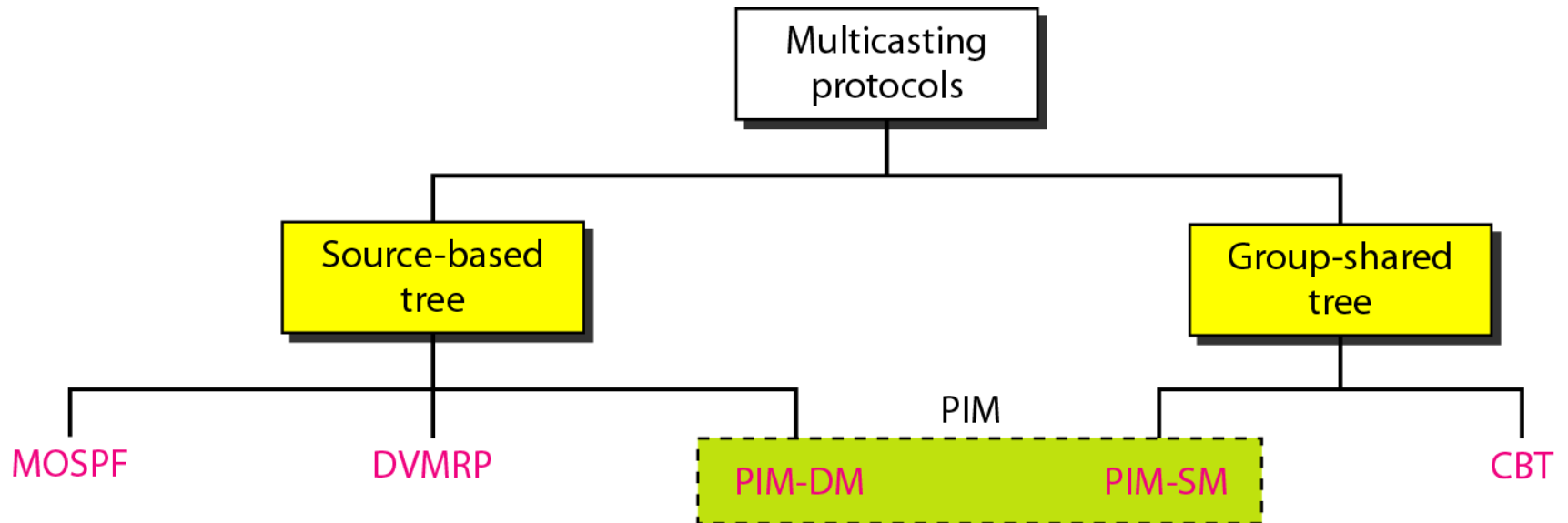
Figure 22.38 *Group-shared tree approach*



Note

In the group-shared tree approach, only the core router, which has a shortest path tree for each group, is involved in multicasting.

Figure 22.39 *Taxonomy of common multicast protocols*





Note

Multicast link state routing uses the source-based tree approach.



Note

**Flooding broadcasts packets, but
creates loops in the systems.**



Note

**RPF eliminates the loop in the
flooding process.**

Figure 22.40 *Reverse path forwarding (RPF)*

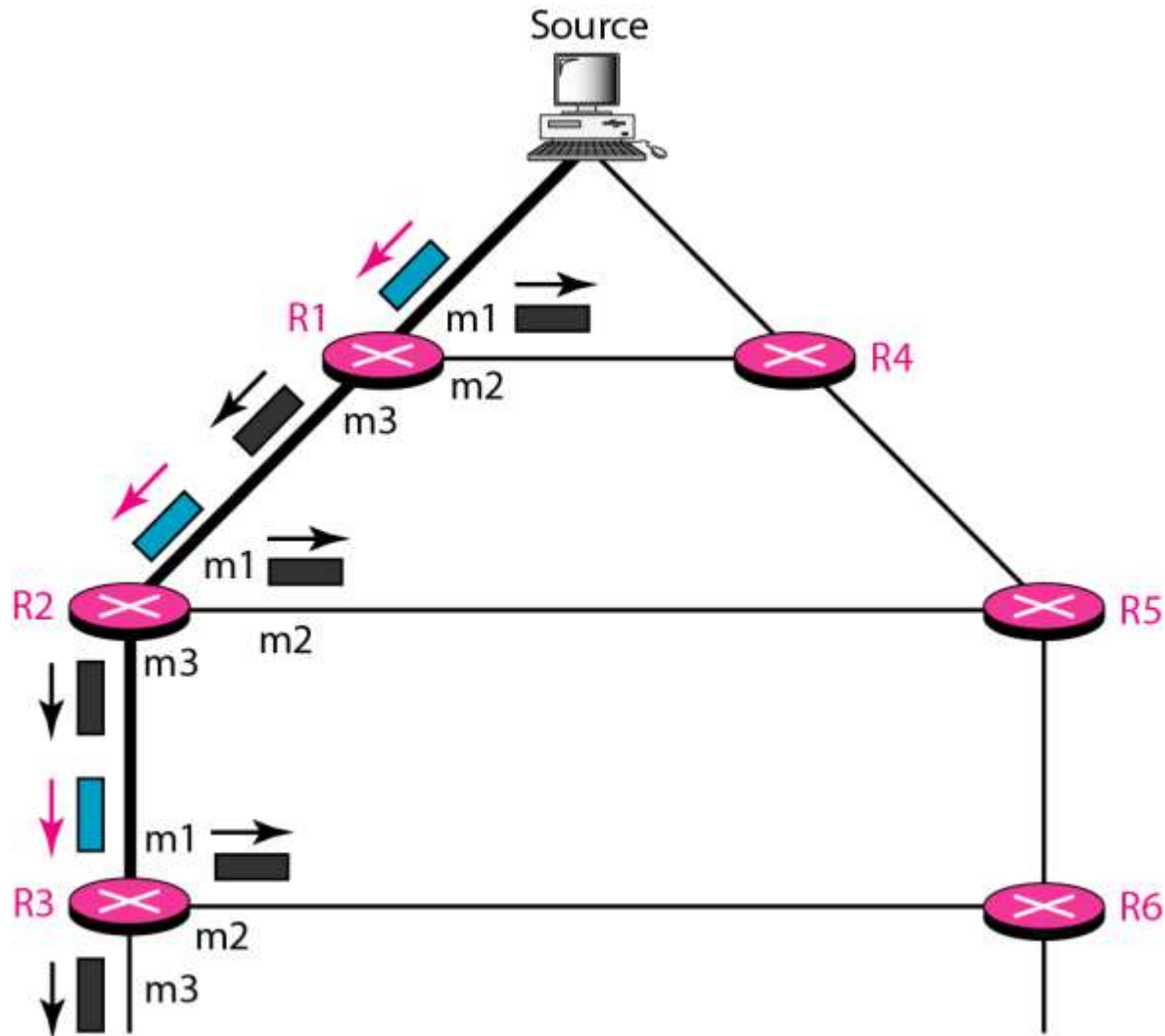


Figure 22.41 *Problem with RPF*

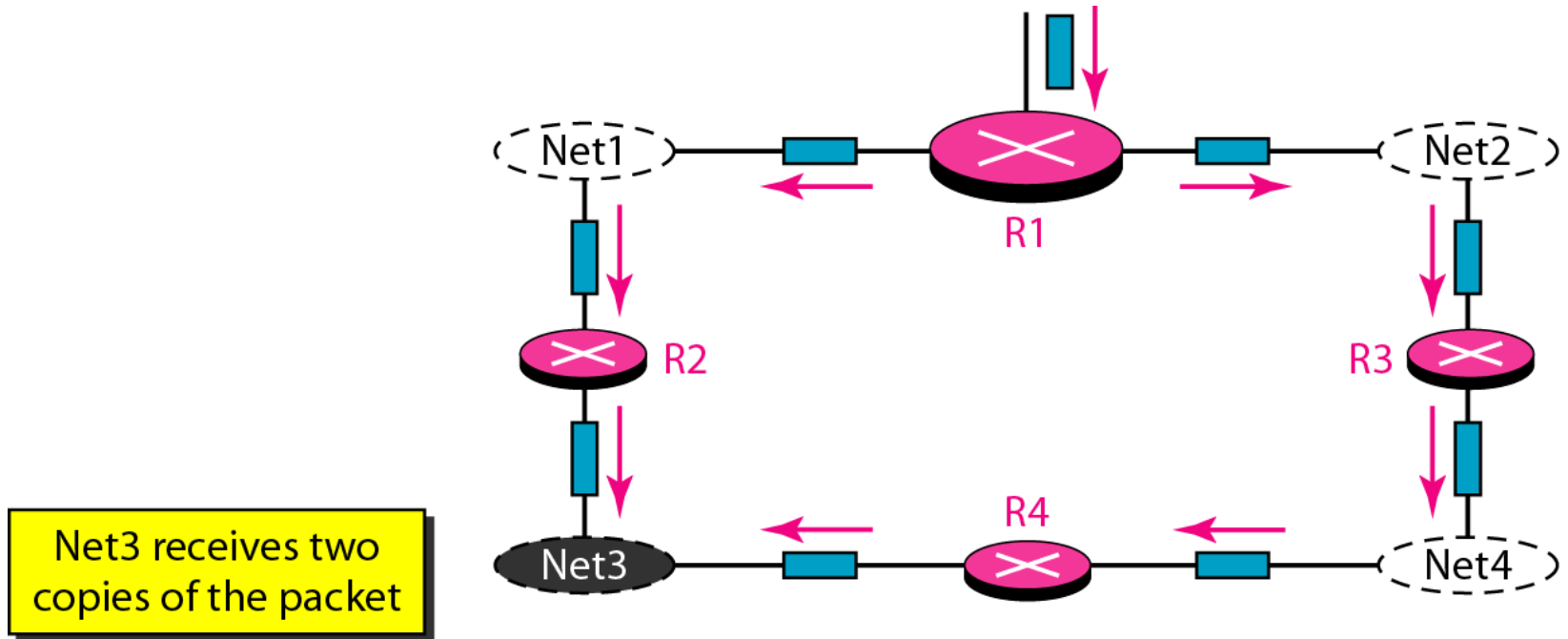
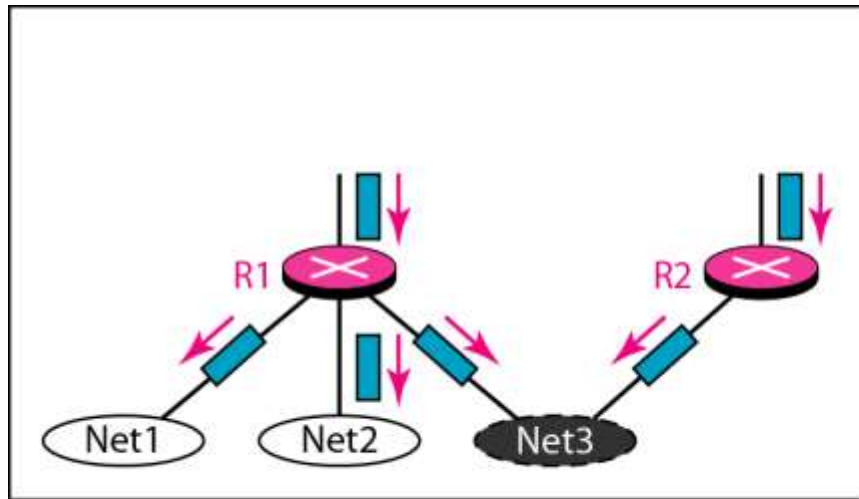
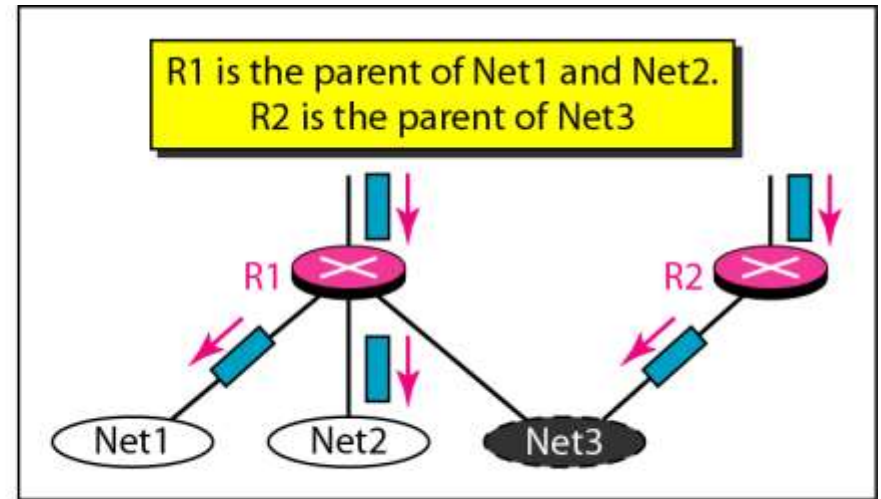


Figure 22.42 *RPF Versus RPB*



a. RPF

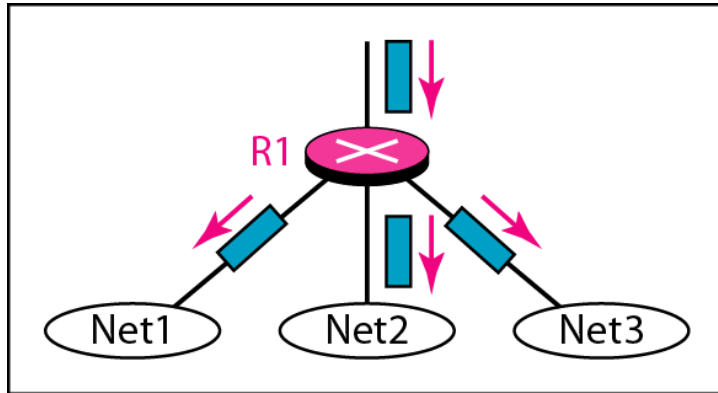


b. RPB

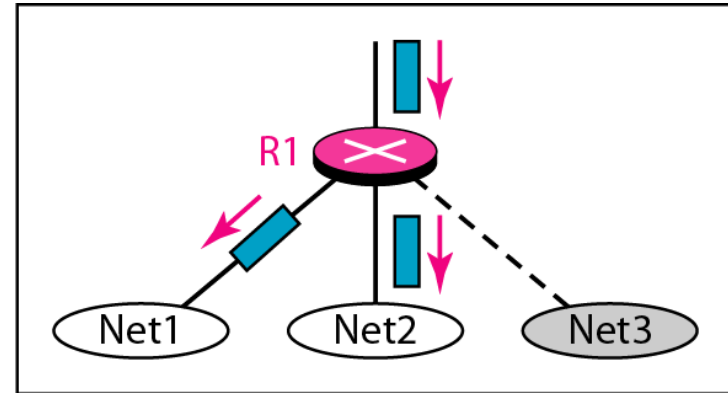
Note

RPB creates a shortest path broadcast tree from the source to each destination. It guarantees that each destination receives one and only one copy of the packet.

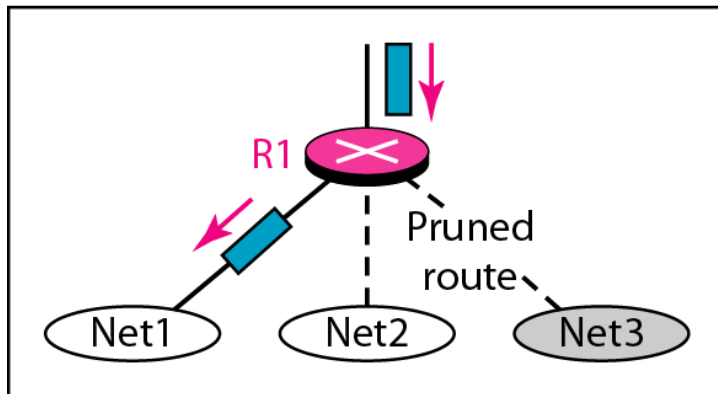
Figure 22.43 *RPF, RPB, and RPM*



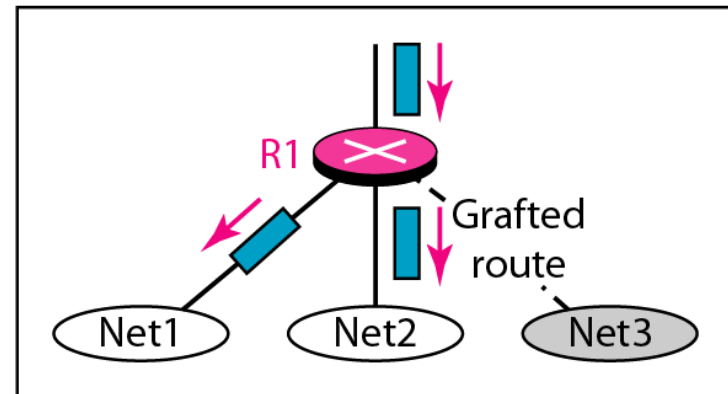
a. RPF



b. RPB



c. RPM (after pruning)



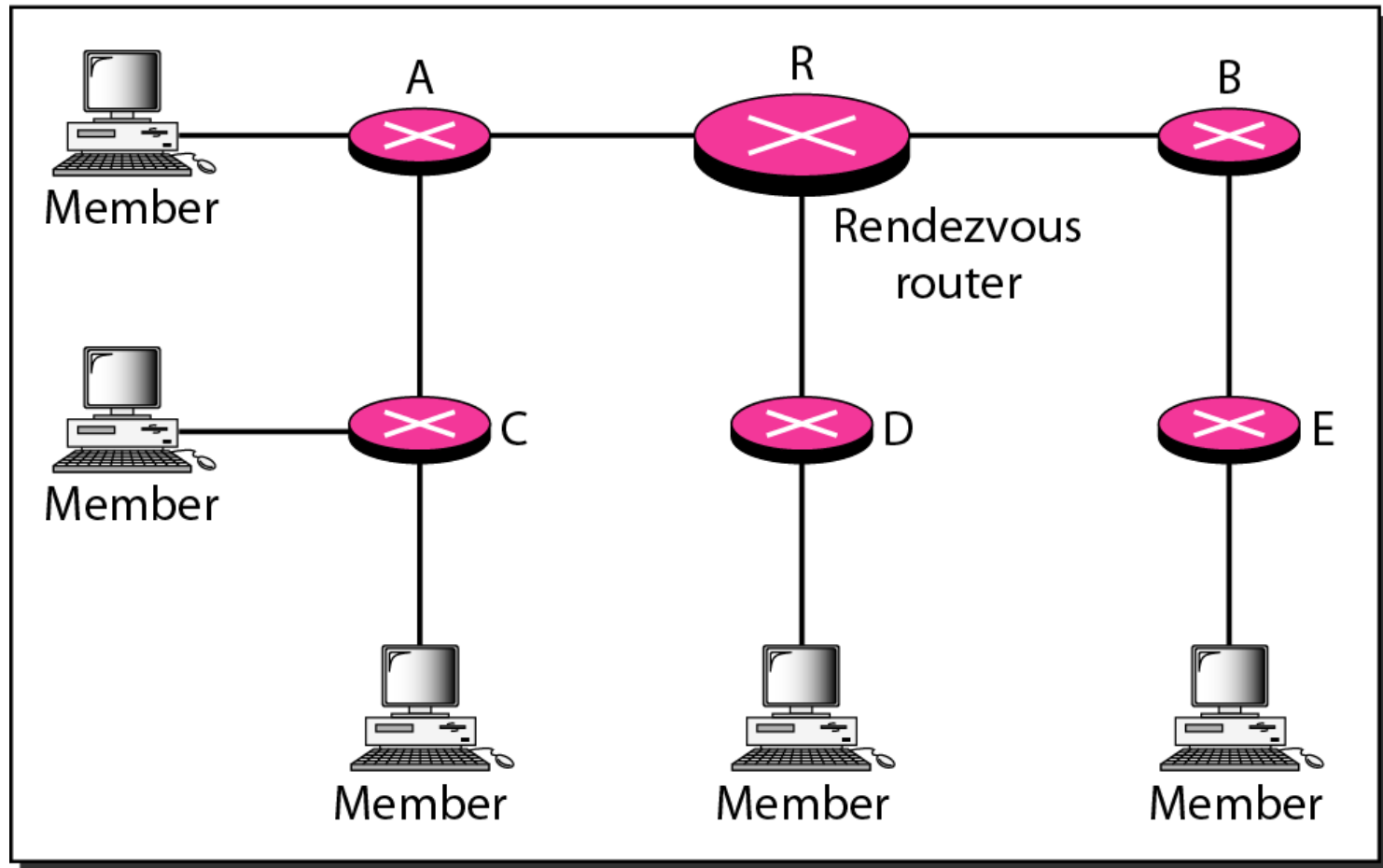
d. RPM (after grafting)

Note

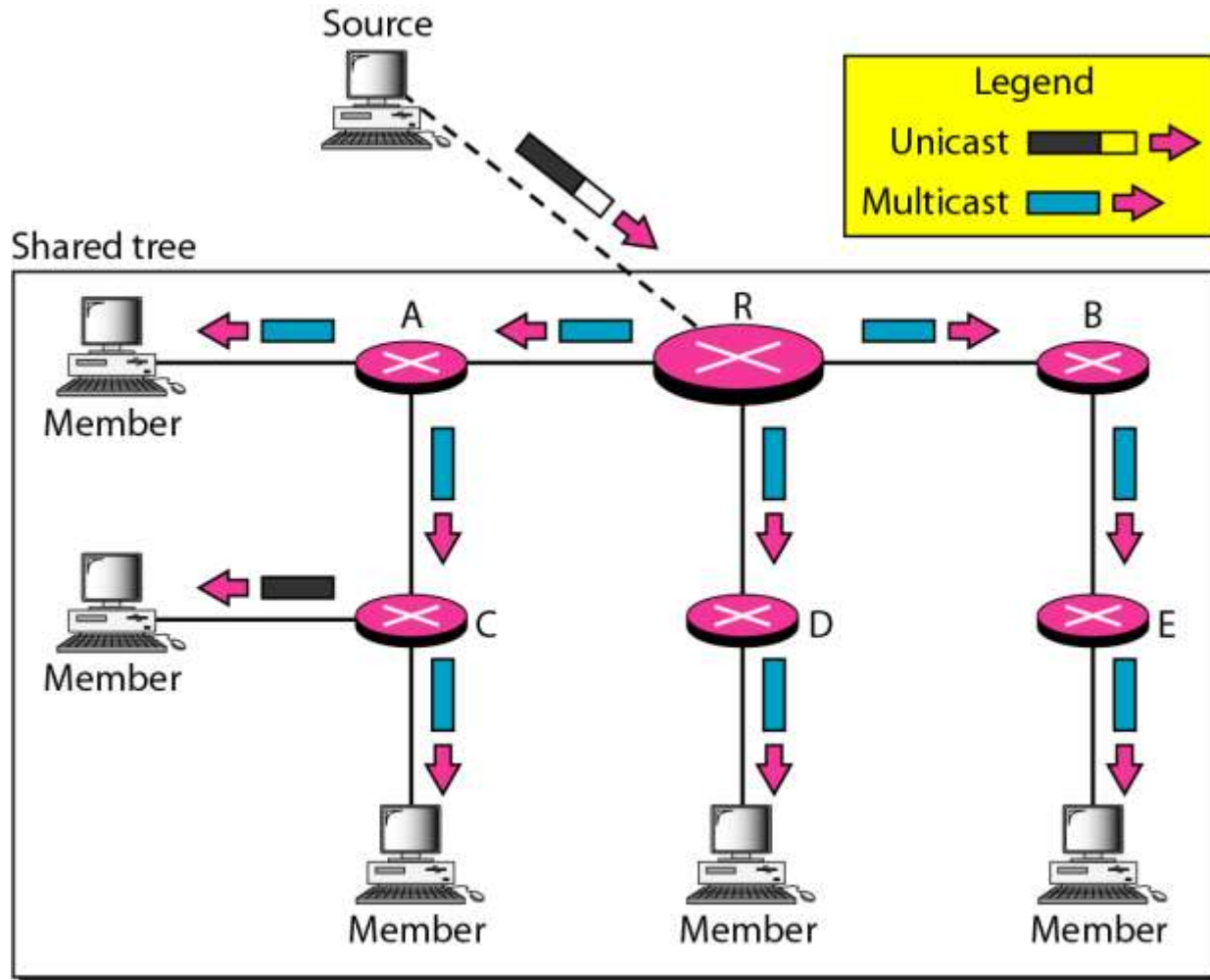
RPM adds pruning and grafting to RPB to create a multicast shortest path tree that supports dynamic membership changes.

Figure 22.44 *Group-shared tree with rendezvous router*

Shared tree



22.60



Note

In CBT, the source sends the multicast packet (encapsulated in a unicast packet) to the core router. The core router decapsulates the packet and forwards it to all interested interfaces.



Note

PIM-DM is used in a dense multicast environment, such as a LAN.

Note

PIM-DM uses RPF and pruning and grafting strategies to handle multicasting.

However, it is independent of the underlying unicast protocol.



Note

PIM-SM is used in a sparse multicast environment such as a WAN.



Note

PIM-SM is similar to CBT but uses a simpler procedure.

Figure 22.46 *Logical tunneling*

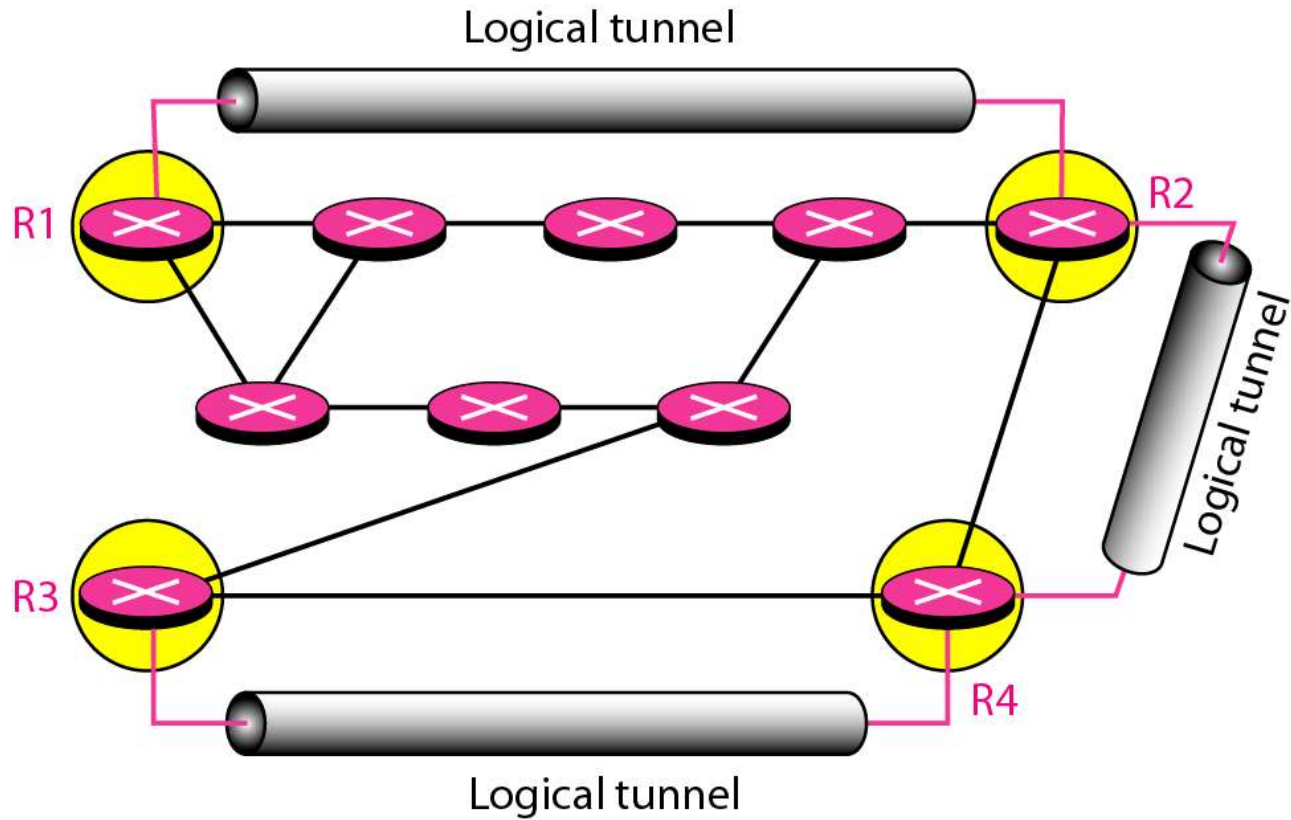


Figure 22.47 *MBONE*

